TCF-Net: An Explainable Tri-Modal Credibility Fusion Network with Selective Abstention for Multilingual Fake News and Hate Speech Detection

A PROJECT REPORT

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ABSTRACT

***Abstract*—**Fake and abusive posts travel quickly across Indian social networks, where a single message may mix English with Tamil or Hindi and switch between native script and Roman letters. Most existing detectors were built for monolingual English news, read only the meaning of a text, and are obliged to give a verdict even when the evidence is weak. This report presents TCF-Net, a Tri-Modal Credibility Fusion Network that addresses all three problems. TCF-Net reads each claim through three evidence streams: (A) contextual semantics from the MuRIL multilingual encoder combined with script-agnostic character n-grams; (B) a 22-dimensional vector of deception-stylometry cues such as hedging, sensational vocabulary, capitalisation and punctuation bursts, built with lexicons for English, Tamil, Tanglish, Hindi and Hinglish; and (C) an 8-dimensional propagation prior obtained from a Susceptible–Infected–Recovered cascade driven by the emotional arousal of the text. A learned gate weighs the three streams for every input, and a shared trunk feeds a veracity head, a SelectiveNet selection head that allows the model to answer "Not Sure", and a jointly trained hate-speech head. On a 5,000-claim, five-language benchmark evaluated with a grouped split that prevents duplicate texts from leaking between training and test data, TCF-Net reaches 92.29% accuracy and a ROC-AUC of 0.9886 when it must answer every claim. With the abstention threshold calibrated on validation data it answers 84.7% of test claims with 100.00% accuracy, deferring every genuinely ambiguous claim to human fact-checkers, and it detects hate speech with 96.50% accuracy. Per-sample modality weights, stylometric profiles and word-level saliency are exposed through an interactive three-tab web application, demonstrated here on native Tamil, Tanglish, Hinglish and English input.

***Index Terms*—**fake news detection, code-mixed text, MuRIL, stylometry, selective prediction, abstention, hate speech, multi-task learning, Tamil, Hinglish, explainable AI.

TABLE OF CONTENTS

I. Introduction 1
A. Background 1
B. Misinformation in the Indian Context 1
C. Motivation 1
D. Problem Statement 1
E. Objectives 1
F. Scope 1
G. Contributions 1
H. Real-World Use Cases 2
I. Organisation of the Report 2
II. Background and Preliminaries 2
A. Transformer Encoders and MuRIL 2
B. Code-Mixing and Transliteration 2
C. Linguistic Cues of Deception 2
D. Epidemic Models of Information Spread 2
E. Selective Prediction and the Reject Option 2
F. Multi-Task Learning 3
G. Evaluation under Duplicated Data 3
III. Literature Review 3
A. Hindi Fake News Detection Using Transformer Ensembles 3
B. Dravidian Fake News Detection with Gradient Accumulation 3
C. Cost-Sensitive Hate Speech Detection in Dravidian Code-Mixed Text 3
D. Survey of Graph Neural Network Methods for Fake News Detection 4
E. Survey of Multimodal Fake News Detection Techniques 4
F. Large Language Models as Advisors for Fake News Detection 4
G. Linguistic Feature Based Word Embedding for Hindi Fake News 5
H. Shared Task on Fake News Detection from Malayalam Social Media 5
I. TELLER: Trustworthy and Explainable Fake News Detection 5
J. Robust Fake News Detection against LLM-Empowered Style Attacks 6
K. ASPEST: Active Selective Prediction under Distribution Shift 6
L. Thematic Synthesis 6
M. Comparative Summary 6
N. Research Gaps 8
IV. Proposed Methodology and System Design 8
A. Design Principles 8
B. System Architecture 8
C. Processing Pipeline 9
D. Confound Scrubbing 9
E. Modality A: Contextual Semantics 9
F. Modality B: Deception Stylometry 9
G. Modality C: SIR Propagation Prior 10
H. Cross-Modal Gated Fusion 11
I. Prediction Heads 11
J. Training Objective 11
K. Risk-Targeted Abstention Calibration 11
L. Inference Workflow 11
M. Model Complexity 12
N. Comparison with Existing Approaches 12
V. Implementation 12
A. Development Environment 12
B. Benchmark Construction 12
C. Dataset Statistics 12
D. Representative Samples 13
E. Leak-Free Grouped Split 13
F. Software Architecture and Modules 13
G. Training Procedure 13
H. Web Application 14

I. Running the System 14
VI. System Demonstration 14
A. Tamil Fake and Hateful Claim 14
B. Tamil Real Claim and Abstention 15
C. Code-Mixed Claims 15
D. English and Hinglish Hate Examples 16
E. Tri-Modal Evidence View 16
F. Risk–Coverage Auditor 17
G. Summary of Demonstration Outputs 17
H. Evidence behind the Demonstration Verdicts 18
VII. Testing and Performance Evaluation 18
A. Testing Strategy 18
B. Unit Testing 18
C. Integration Testing 18
D. Functional Test Cases 18
E. Evaluation Metrics 18
VIII. Performance Evaluation and Discussion 19
A. Overall Performance 19
B. Training Behaviour 19
C. Confusion Analysis 19
D. ROC and Precision–Recall Analysis 20
E. Selective Prediction 20
F. Confidence Distribution 21
G. Performance by Claim Type and Language 21
H. Modality Gate Analysis 22
I. Stylometric Evidence 22
J. Propagation Prior by Language 23
K. Language by Claim-Type Breakdown 23
L. Hate-Speech Analysis 23
M. Threshold Sensitivity 24
N. Efficiency 24
O. Baselines and Ablation 24
P. Error Analysis 25
Q. Discussion 25
R. Changes since Review II 25
S. Lessons Learned 25
T. Threats to Validity 26
IX. Ethical Considerations 26
A. Human Oversight 26
B. Fairness across Languages and Communities 26
C. Risk of Misuse and Over-Reliance 26
D. Data and Privacy 26
X. Conclusion and Future Enhancements 26
A. Conclusion 26
B. Limitations 26
C. Deployment Roadmap 26
D. Future Enhancements 27
Appendix 1: Coding 27
Appendix 2: Sample Benchmark Records 30
Appendix 3: Sample Test Predictions 30
Appendix 4: Complete Multilingual Lexicons 31
Appendix 5: Benchmark Template Pools 31
Appendix 6: Glossary 32
Appendix 7: User Manual 32
Appendix 8: Reproducibility Details 32
Appendix 9: Additional Screenshots 32
References 35

LIST OF TABLES

TABLE I Literature review summary (11 works, 2023–2024) 7
TABLE II Comparison of reviewed works with TCF-Net 8
TABLE III Datasets in the reviewed literature and this work 8
TABLE IV The 22 stylometric deception features 10
TABLE V Multilingual lexicons used by the stylometric stream 10
TABLE VI The eight SIR propagation descriptors 10
TABLE VII Trainable parameters of TCF-Net 12
TABLE VIII Existing approaches versus the proposed system 12
TABLE IX Development environment 12
TABLE X Composition of the five-language benchmark 12
TABLE XI Representative claims from each language 13
TABLE XII Data split produced by the grouped protocol 13
TABLE XIII Source modules and responsibilities 13
TABLE XIV Training hyper-parameters 14
TABLE XV User-interface components of the web application 14
TABLE XVI Artifacts produced by the training pipeline 14
TABLE XVII Model outputs for the demonstration claims 17
TABLE XVIII Stylometric and propagation evidence for the demonstration claims 18
TABLE XIX Functional and integration test cases 18
TABLE XX Test-set performance of TCF-Net 19
TABLE XXI Training history (per epoch) 19
TABLE XXII Selective error at representative coverage levels 21
TABLE XXIII Accuracy and abstention by claim type and language 21
TABLE XXIV Test results by language and claim type 23
TABLE XXV Hate-speech precision, recall and F1 by language 24
TABLE XXVI Coverage and accuracy as a function of the threshold τ 24
TABLE XXVII Model size and measured CPU latency 24
TABLE XXVIII Comparison with baselines and ablations on the same split (100% coverage) 24
TABLE XXIX Example full-coverage errors (all abstained at τ) 25
TABLE XXX Earlier checkpoint versus final model 25
TABLE XXXI Sample benchmark records 30
TABLE XXXII Sample test predictions 30
TABLE XXXIII Complete lexicon entries 31
TABLE XXXIV Template pools used to generate the benchmark 31
TABLE XXXV Glossary of key terms 32

LIST OF FIGURES

Fig. 1. System architecture of TCF-Net. 9
Fig. 2. End-to-end processing pipeline from input claim to decision. 9
Fig. 3. SIR propagation prior for a sensational versus a neutral claim. 11
Fig. 4. Label distribution by language and ambiguity level. 13
Fig. 5. Tab 1 – Tamil fake and hateful claim: FAKE + HATEFUL. 15
Fig. 6. Tab 1 – Tamil real claim and abstention on an unverifiable claim. 15
Fig. 7. Tab 1 – Tanglish and Hinglish claims. 16
Fig. 8. Tab 1 – English and Hinglish claims. 16
Fig. 9. Tab 2 – stylometric profile and SIR cascade for the Tamil hateful claim. 17
Fig. 10. Tab 3 – risk–coverage curve and abstention audit tables. 17
Fig. 11. Training behaviour and per-language performance. 19
Fig. 12. Confusion matrices for veracity and hate speech on the test set. 20
Fig. 13. Veracity confusion matrices by language. 20
Fig. 14. ROC and precision–recall curves for veracity and hate speech. 20
Fig. 15. Risk–coverage curve of TCF-Net compared with random abstention. 21
Fig. 16. Distribution of g(x) by claim type, and P(fake) versus g(x). 21
Fig. 17. Mean modality gate weights by language and by claim type. 22
Fig. 18. Stylometric feature distributions by claim type. 22

Fig. 19. Mean gate weights and the most discriminative stylometric cues. 23
Fig. 20. Mean SIR propagation prior by language and veracity class. 23
Fig. 21. Test accuracy of baselines, ablations and the full model. 25
Fig. 22. Tab 1 on a Hinglish fake and hateful claim (full page). 33
Fig. 23. Tab 2 on a Hinglish claim (full page). 34
Fig. 24. Tab 3 risk–coverage auditor (full page). 35

LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Symbol	Meaning	Symbol	Meaning
AMP	Automatic Mixed Precision	SVD	Singular Value Decomposition
AUC / ROC-AUC	Area Under the Receiver Operating Characteristic Curve	TCF-Net	Tri-Modal Credibility Fusion Network (proposed)
AURC	Area Under the Risk–Coverage Curve	TF-IDF	Term Frequency–Inverse Document Frequency
BERT	Bidirectional Encoder Representations from Transformers	Tanglish / Hinglish	Tamil–English / Hindi–English code-mixed text in Roman script
CE	Cross-Entropy loss	x	Input claim
DFND	Dravidian Fake News Dataset	z_t, z_s, z_p	Projected semantic, stylometric and propagation representations
EPJ	Embedded Project (course component)	α (alpha)	Per-sample modality gate weights, $\alpha \in \mathbb{R}^3$, $\Sigma\alpha = 1$
F1	Harmonic mean of precision and recall	f(x)	Veracity prediction head (Real / Fake)
GNN	Graph Neural Network	g(x)	Selection (confidence) head, $g(x) \in [0, 1]$
HASOC	Hate Speech and Offensive Content Identification (FIRE shared task)	h(x)	Auxiliary veracity head
LLM	Large Language Model	τ (tau)	Abstention threshold on g(x)
MLP	Multi-Layer Perceptron	c	Target coverage
MuRIL	Multilingual Representations for Indian Languages	φ (phi)	Empirical coverage
NLP	Natural Language Processing	β, γ	SIR transmission and recovery rates
SIR	Susceptible–Infected–Recovered epidemic model	λ (lambda)	Coverage-penalty weight

I. Introduction

A. Background

Social media has become a main route to news for a large share of Indian internet users. WhatsApp, X, Facebook and YouTube let one person reach thousands of others within minutes, and the same openness makes these platforms efficient carriers of rumours, fabricated stories and inflammatory speech. Large-scale studies of online diffusion found that false stories spread farther, faster and more broadly than true ones, largely because they are novel and emotionally charged [1]. During elections, disease outbreaks and communal tensions, such content has led to panic buying, vaccine hesitancy and even mob violence.

Automatic fake-news detection is the task of judging, from a post and its context, whether the claim it carries is credible [2]. Early systems relied on hand-crafted lexical features and classical classifiers [3], and benchmark collections such as LIAR made the task measurable [4]. Later work adopted pre-trained transformers such as BERT [5] and graph neural networks that model how stories propagate between users [6]. In parallel, hate-speech detection identifies content that attacks people for their religion, region, caste or language, and has its own shared tasks for Indian languages [7], [8].

B. Misinformation in the Indian Context

India adds difficulties that most published systems do not face. More than twenty scheduled languages are written in a dozen scripts, and online users routinely mix two languages inside one sentence. A Tamil speaker may type "VIRAL ALERT: semester exam schedule full scam nu report leak aagirukku", which combines English nouns, Tamil grammar and Roman spelling. The same claim may circulate in native Tamil script, in transliterated "Tanglish", and in English translation. Hindi and Hinglish show the same pattern. Detectors trained on English news, or on native-script text alone, therefore see an unfamiliar distribution when deployed on Indian social media.

Misinformation in this setting is also tightly bound to hate speech. Rumours about a community are often accompanied by calls to drive its members out, and a single forwarded message can be false and hateful at the same time. Treating the two problems with separate models doubles the engineering effort and ignores the emotional, aggressive language they share.

C. Motivation

Three observations motivated this project. First, deceptive writing has a recognisable style: it hedges, exaggerates, shouts in capital letters and urges readers to share before the content is "deleted". Psychological studies of deception have linked such linguistic markers to lying [9], and they are cheap to measure in any language, yet most neural detectors ignore them. Second, misinformation is a social phenomenon: how a message is likely to spread is evidence in its own right, even before real sharing data exist. Third, and most important for deployment, a detector that must label every post will make confident mistakes on claims that simply cannot be verified from the text. Fact-checking organisations would rather receive a short list of uncertain items for manual review than a stream of wrong automatic verdicts.

A practical credibility assistant for Indian social media should therefore combine several kinds of evidence, handle code-mixed text in several scripts, explain why it reached a decision, screen for hate speech at the same time, and know when not to answer. TCF-Net was designed around exactly these requirements.

D. Problem Statement

Given a short social-media claim x written in English, Tamil, Tamil–English (Tanglish), Hindi or Hindi–English (Hinglish), the system must (i) decide whether the claim is real or fake; (ii) decide whether it contains hate speech; (iii) produce a confidence score g(x) and return "Not Sure" when g(x) falls below a calibrated threshold; and (iv) explain the decision in terms of the evidence streams and words that drove it. The system must be evaluated so that no information from the test data reaches training or calibration.

E. Objectives

1. Design a tri-modal network that fuses contextual semantics, deception stylometry and a simulated propagation signal through a learned, per-sample gate.
2. Support code-mixed and native-script input by pairing the MuRIL encoder with script-agnostic character n-grams and lexicons for five language settings.
3. Integrate a SelectiveNet abstention mechanism so the model can defer uncertain claims to human fact-checkers at a chosen operating point.
4. Train a hate-speech head jointly with the veracity task so that one model screens for both harms.
5. Evaluate rigorously with a leak-free grouped split, classical baselines, modality ablations, per-language analysis and risk–coverage curves.
6. Deliver an interactive web application that exposes verdicts, confidence, modality weights and word-level saliency for every input, including native Tamil text.

F. Scope

The project handles short claims of up to 128 sub-word tokens, typical of forwarded messages, tweets and comments. It uses text-only evidence; images, video, audio and retrieval of external evidence are outside the present scope and are discussed as future work. The propagation stream is simulated from the text and needs no platform data, so the system can be used at the moment a post first appears.

G. Contributions

- **Tri-modal credibility fusion** with an entropy-regularised gate that prevents the network from collapsing onto a single evidence stream.
- **Multilingual deception stylometry**: a 22-feature extractor with a script-aware tokenizer and lexicons for English, Tamil, Tanglish, Hindi and Hinglish.

- **Label-free propagation prior**: a 24-step SIR cascade parameterised only by textual arousal, which adds virality evidence without any risk of label leakage.
- **Selective multi-task learning**: SelectiveNet abstention for veracity trained jointly with hate-speech detection, with a risk-targeted threshold calibrated on validation data.
- **Leak-free evaluation**: grouped splits, five classical baselines, a modality ablation and analyses by language, claim type and modality.
- **Explainable demonstration**: a three-tab web application shown end to end on Tamil, Tanglish, Hinglish and English input.

H. Real-World Use Cases

TCF-Net is designed as decision support for people who handle large volumes of forwarded content. Four settings illustrate its intended use.

- **Fact-checking desks.** Incoming tips are triaged automatically: clear fabrications and clear official notices are labelled with evidence, and the "Not Sure" queue becomes the reviewers' work list.
- **Messaging tip lines.** Users forward a suspicious message to a bot and receive a verdict, the cues behind it and, when the system abstains, a note that the claim has been passed to a human checker.
- **Platform moderation.** Posts flagged as both fake and hateful can be prioritised for review during elections or communal tension, when speed matters most.
- **Newsrooms and public agencies.** Journalists and officials can screen viral claims in Tamil, Hindi or code-mixed form before responding, using the saliency view to see which words drive the verdict.

I. Organisation of the Report

Section II introduces the background concepts used throughout. Section III reviews eleven recent works and identifies research gaps. Section IV presents the proposed methodology and system design. Section V describes the implementation and dataset, and Section VI demonstrates the working system. Section VII covers testing and the evaluation metrics, Section VIII evaluates performance and discusses the results, Section IX considers ethical issues and Section X gives the conclusion and future enhancements. Appendix 1 lists the core code, and the remaining appendices give sample records and predictions, lexicons, templates, a glossary, a user manual, reproducibility details and additional screenshots.

II. Background and Preliminaries

A. Transformer Encoders and MuRIL

Transformers process a sequence of sub-word tokens with stacked self-attention layers that let every token attend to every other token [10]. BERT pre-trains such an encoder with masked-language modelling on large unlabelled corpora and fine-tunes it for downstream tasks [5]; DistilBERT compresses BERT into a smaller student model [11]. MuRIL applies the same recipe to Indian languages: it is pre-trained on 17 Indian languages together with their transliterated forms and parallel translations, which makes it well suited to text that mixes scripts [12]. In TCF-Net the encoder is kept frozen and its final hidden states are mean-pooled into a single sentence vector.

$$e = \Sigma_j \, m_j \, h_j \, / \, \Sigma_j \, m_j \qquad (1)$$

Here h_j is the final hidden state of token j and $m_j \in \{0, 1\}$ is its attention-mask value, so padding tokens do not contribute.

B. Code-Mixing and Transliteration

Code-mixing is the alternation of two languages within one utterance, and transliteration is writing a language in another script, usually Roman letters. Both break assumptions of standard tokenisers. Romanised Tamil and Hindi have no standard spelling ("paazhadikiranga", "pazhadikiraanga"), and native-script words contain combining vowel signs and viramas that generic word tokenisers split apart. Character n-grams are robust to both problems because they match recurring sub-strings regardless of spelling variants or word boundaries.

C. Linguistic Cues of Deception

Research on deceptive language has found that liars tend to use fewer self-references, more negative emotion words and simpler cognitive structure [9]. Online misinformation adds its own markers: sensational vocabulary ("SHOCKING", "EXPOSED"), urgency ("share before it is deleted"), capitalised words, clusters of exclamation marks and appeals to hidden authority ("sources say"). Readability indices such as Flesch reading ease [13] summarise sentence and word complexity. Such cues are not proof of falsehood, but they shift the probability that a message is deceptive and are largely language-independent once suitable lexicons exist.

D. Epidemic Models of Information Spread

Compartmental epidemic models divide a population into susceptible (S), infected (I) and recovered (R) groups [14]. Rumour-spreading models adapt the same idea: people who have not yet seen a message are susceptible, active sharers are infected, and those who have lost interest are recovered [15]. The transmission rate β controls how quickly a message spreads and the recovery rate γ how quickly interest fades. Emotionally arousing messages behave like a more contagious disease: higher β, lower γ, a taller and earlier peak.

E. Selective Prediction and the Reject Option

A selective classifier may abstain on an input instead of predicting a label. The idea dates back to Chow's analysis of the optimal trade-off between error and rejection [16]. Its quality is described by two numbers: coverage, the fraction of inputs it answers, and selective risk, its error rate on those answered inputs. SelectiveNet trains a dedicated selection head g(x) jointly with the classifier and penalises deviations from a target coverage [17]. Lowering coverage should lower risk; a good selector lowers it much faster than random abstention.

F. Multi-Task Learning

Multi-task learning trains one model on several related tasks with shared representations, so that signals useful for one task regularise the others [18]. Veracity and hate speech share emotive, aggressive language, so a shared trunk with task-specific heads is a natural design. Samples annotated for only one task can still be used by masking the loss of the missing task.

G. Evaluation under Duplicated Data

When a corpus contains repeated or templated texts, a random split places copies of the same sentence in both training and test sets, so a model can score highly by memorisation. Grouped cross-validation assigns all copies of a group to the same fold. TCF-Net groups samples by their normalised text and stratifies by veracity and language, so the test set contains only sentences the model has never seen.

III. Literature Review

Eleven recent works published between 2023 and 2024 by Elsevier, Springer, IEEE, ACM, AAAI, ACL and TMLR were reviewed. They cover four threads that TCF-Net brings together: transformer-based detection for Indian languages (Hindi, Dravidian and Malayalam), hate speech in code-mixed text, explainable and style-robust detection, and selective prediction. For each work the review records (1) the title, authors, date and publisher, (2) the problem and objectives, (3) the methodology adopted and (4) the performance and limitations.

A. Hindi Fake News Detection Using Transformer Ensembles

1) Title, Authors, Date and Publisher

"Hindi fake news detection using transformer ensembles", by A. Praseed, J. Rodrigues and P. S. Thilagam; published 2023, vol. 119, Art. no. 105731 in Engineering Applications of Artificial Intelligence, Elsevier [19].

2) Problem and Objectives

Hindi is one of the most widely spoken languages in the world, yet labelled resources and models for detecting fake Hindi news lag far behind those for English. The authors aim to build an accurate Hindi fake-news classifier that works under the data constraints typical of low-resource languages, where large annotated corpora are not available and single models over-fit easily.

3) Methodology Adopted

Several pre-trained multilingual transformers, namely XLM-RoBERTa, multilingual BERT and ELECTRA, are fine-tuned independently on Hindi news. Their predictions are then combined in an ensemble so that the partly different errors of each member cancel out. The ensemble is compared against each individual fine-tuned model on the same data.

4) Performance and Limitations

The ensemble is more accurate than any single member, but running three full transformers for every article multiplies latency and memory, which is a concern for real-time screening of social media. The study covers Devanagari Hindi only, so Romanised Hinglish posts, which dominate informal platforms, are not addressed. Every article receives a label; there is no confidence-based deferral and no explanation of which cues drove a decision.

5) Relevance to TCF-Net

Shows that multilingual transformers transfer to Hindi; TCF-Net keeps a single frozen encoder instead of an ensemble and adds Romanised Hinglish and abstention.

B. Dravidian Fake News Detection with Gradient Accumulation

1) Title, Authors, Date and Publisher

"Dravidian fake news detection with gradient accumulation based transformer model", by E. Raja, B. Soni, S. K. Borgohain and C. Lalrempuii; published December 2023, pp. 466–471 in Proc. 20th International Conference on Natural Language Processing (ICON 2023), ACL Anthology [20].

2) Problem and Objectives

Tamil, Telugu, Kannada and Malayalam have little annotated data for fake-news detection, and fine-tuning large models on them is limited by GPU memory. The goal is to detect fake news across these Dravidian languages accurately while keeping training stable on modest hardware.

3) Methodology Adopted

The MuRIL transformer, pre-trained on Indian languages, is fine-tuned with gradient accumulation, which simulates large batches inside a small memory budget, together with adaptive learning-rate scheduling. The work is associated with the Dravidian Fake News Dataset (DFND), released on IEEE DataPort with more than 27,000 articles balanced between fake and real across four languages.

4) Performance and Limitations

The study confirms that MuRIL is a strong backbone for Dravidian text, a finding TCF-Net builds on. It relies on semantics alone and ignores stylistic and propagation cues. The data are native-script news articles, so behaviour on short Romanised code-mixed posts is untested, and the classifier must label every article.

5) Relevance to TCF-Net

Directly motivates the MuRIL backbone of TCF-Net, which extends it from native-script articles to short code-mixed posts and adds non-semantic evidence.

C. Cost-Sensitive Hate Speech Detection in Dravidian Code-Mixed Text

1) Title, Authors, Date and Publisher

"Detection of hate speech and offensive language CodeMix text in Dravidian languages using cost-sensitive learning approach", by K. Sreelakshmi, B. Premjith, B. R. Chakravarthi and K. P. Soman; published 2024, vol. 12 in IEEE Access, Institute of Electrical and Electronics Engineers [21].

2) Problem and Objectives

Hate and offensive speech in Kannada–English, Malayalam–English and Tamil–English code-mixed posts is hard to detect because the text is noisy and transliterated, and the classes are heavily imbalanced, with far fewer offensive than benign posts. The objective is a classifier that copes with this imbalance across several Dravidian code-mixed datasets.

3) Methodology Adopted

Posts are represented with multilingual transformer embeddings that are classified by classical machine learning models. Cost-sensitive learning gives the minority (offensive) class a higher misclassification cost so that the classifier does not collapse onto the majority class. Six openly available code-mixed datasets, including HASOC and DravidianLangTech corpora, are used for evaluation.

4) Performance and Limitations

Reported accuracy on the Tamil code-mixed sets is in the seventies, which shows how hard real code-mixed hate detection remains. The embeddings are used as fixed features rather than fine-tuned, and the system addresses offensive language only: a separate model would be needed to judge veracity. It has no reject option for uncertain posts.

5) Relevance to TCF-Net

Defines the hate-speech side of the problem in Dravidian code-mixed text; TCF-Net handles it with a joint head instead of a separate model.

D. Survey of Graph Neural Network Methods for Fake News Detection

1) Title, Authors, Date and Publisher

"Fake news detection: A survey of graph neural network methods", by H. T. Phan, N. T. Nguyen and D. Hwang; published May 2023, vol. 139, Art. no. 110235 in Applied Soft Computing, Elsevier [6].

2) Problem and Objectives

Fake news is shaped not only by its content but by who shares it and how it spreads. Graph neural networks (GNNs) can model these relations, but the literature applying them was fragmented. The survey organises GNN-based detection, explains how such systems are built and identifies open problems.

3) Methodology Adopted

The authors review the state of the art, outline a general procedure for building a GNN-based detector, and propose a taxonomy that groups models by the family of graph neural network they use. For each group they analyse representative models, the graphs they construct (news–user graphs, propagation trees and heterogeneous graphs) and the datasets used.

4) Performance and Limitations

The survey shows that propagation structure is highly informative, which motivates the propagation stream of TCF-Net. It also exposes a practical weakness: GNN methods need sharing graphs and user data that are often private, unavailable or not yet formed when a post first appears, and most benchmarks are English.

5) Relevance to TCF-Net

Motivates the propagation stream, which TCF-Net approximates with a text-driven SIR prior because sharing graphs are unavailable at posting time.

E. Survey of Multimodal Fake News Detection Techniques

1) Title, Authors, Date and Publisher

"A comprehensive survey of multimodal fake news detection techniques: advances, challenges, and opportunities", by S. Tufchi, A. Yadav and T. Ahmed; published 2023, vol. 12 in International Journal of Multimedia Information Retrieval, Springer [22].

2) Problem and Objectives

Modern misinformation combines text with images and other media, and single-modality detectors miss inconsistencies between them. The survey reviews multimodal detection techniques and identifies challenges and opportunities for research.

3) Methodology Adopted

The authors systematically review multimodal detectors, compare classification models across text- and image-based fake-news datasets, and discuss feature extraction and fusion strategies such as early, late and attention-based fusion before outlining open challenges.

4) Performance and Limitations

The survey makes a strong case that fusing complementary evidence outperforms any single modality, which is the principle behind the gated fusion of TCF-Net. Its focus, however, is image–text fusion on largely English and Chinese benchmarks; it gives little attention to code-mixed Indian languages, to stylometric evidence as a modality, or to selective prediction.

5) Relevance to TCF-Net

Supports the evidence-fusion principle; TCF-Net applies it to three text-derived streams with a learned, per-sample gate.

F. Large Language Models as Advisors for Fake News Detection

1) Title, Authors, Date and Publisher

"Bad actor, good advisor: Exploring the role of large language models in fake news detection", by B. Hu, Q. Sheng, J. Cao, Y. Shi, Y. Li, D. Wang and P. Qi; published March 2024, vol. 38, no. 20, pp. 22105–22113 in Proceedings of the AAAI Conference on Artificial Intelligence, AAAI Press [23].

2) Problem and Objectives

Large language models (LLMs) such as GPT-3.5 have broad world knowledge, but it was unclear whether they detect fake news better than smaller fine-tuned models. The paper asks this question and investigates how LLMs can be used effectively for the task.

3) Methodology Adopted

An empirical study first compares GPT-3.5 with a fine-tuned BERT detector. The authors then propose the Adaptive Rationale Guidance (ARG) network, in which a small language model selectively absorbs multi-perspective

rationales generated by the LLM, and a distilled variant, ARG-D, that removes the need to query the LLM at inference time.

4) Performance and Limitations

GPT-3.5 alone still trails a fine-tuned BERT model, while ARG and ARG-D outperform small-model, LLM-only and combined baselines on two real-world datasets. Querying an LLM adds cost and latency, the evaluation covers English and Chinese only, and the system still forces a decision on every item.

5) Relevance to TCF-Net

Suggests LLM rationales as a future advisory stream; TCF-Net stays LLM-free so that it runs on a CPU in under 100 ms.

G. Linguistic Feature Based Word Embedding for Hindi Fake News

1) Title, Authors, Date and Publisher

"LFWE: Linguistic feature based word embedding for Hindi fake news detection", by R. Sharma and A. Arya; published June 2023, vol. 22, no. 6 in ACM Transactions on Asian and Low-Resource Language Information Processing, ACM [24].

2) Problem and Objectives

Hindi fake-news research lacked a sizeable annotated corpus and representations that capture the linguistic character of deceptive Hindi text. The objectives are to build such a corpus and a detector that exploits linguistic features alongside word embeddings.

3) Methodology Adopted

The authors introduce the HinFakeNews dataset of roughly 33,300 annotated Hindi articles and a linguistic-feature-based word embedding (LFWE) that enriches word representations with hand-crafted linguistic features before classification. They report an accuracy of 98.49% on their data.

4) Performance and Limitations

The corpus contains long-form Devanagari news, so Romanised and code-mixed social-media posts are not covered. The work treats veracity in isolation, with no hate-speech screening and no mechanism to abstain on claims that cannot be verified. The very high accuracy also suggests a relatively separable dataset, a caution that applies to template-based benchmarks too.

5) Relevance to TCF-Net

Confirms the value of linguistic features for Hindi; TCF-Net generalises the idea to five language settings through multilingual lexicons.

H. Shared Task on Fake News Detection from Malayalam Social Media

1) Title, Authors, Date and Publisher

"Overview of the shared task on fake news detection from social media text", by M. Subramanian et al.; published September 2023, pp. 59–63 in Proc. 3rd Workshop on Speech and Language Technologies for Dravidian Languages (DravidianLangTech), RANLP 2023 [25].

2) Problem and Objectives

Fake-news detection for Dravidian social media lacked a shared benchmark. The task asks participants to classify Malayalam social-media posts and comments as fake or original, so that methods can be compared on common data.

3) Methodology Adopted

The organisers collected 5,091 posts and comments from YouTube, split them into training, validation and test sets, and evaluated systems with macro-F1. Eight teams participated; most fine-tuned multilingual transformers such as XLM-RoBERTa and multilingual BERT, and some used classical machine learning.

4) Performance and Limitations

The best systems reached macro-F1 scores only in the 0.8s, which shows that real Dravidian social-media text remains difficult. The benchmark covers a single language and all submitted systems are forced classifiers without abstention or explanation.

5) Relevance to TCF-Net

Shows how hard real Dravidian social-media text is, which is why TCF-Net reports its synthetic-benchmark results with caution.

I. TELLER: Trustworthy and Explainable Fake News Detection

1) Title, Authors, Date and Publisher

"TELLER: A trustworthy framework for explainable, generalizable and controllable fake news detection", by H. Liu, W. Wang, H. Li and H. Li; published August 2024, pp. 15556–15583 in Findings of the Association for Computational Linguistics: ACL 2024 [26].

2) Problem and Objectives

Neural fake-news detectors are opaque, generalise poorly to new domains and are hard to correct. The authors aim for detection that is explainable, generalisable and controllable by human experts.

3) Methodology Adopted

TELLER uses a dual-system design. A cognition system turns expert-written logical predicates into questions that an LLM answers, producing human-readable logic atoms. A decision system then learns generalisable logic rules that aggregate these atoms into a verdict, so experts can inspect and edit the rules.

4) Performance and Limitations

The framework depends on repeated LLM queries, which adds cost and latency, and its benchmarks are English. It explains decisions but still outputs a verdict for every claim, and it does not address code-mixed text or hate speech.

5) Relevance to TCF-Net

Shares the goal of explainability; TCF-Net explains through gate weights, stylometric profiles and word saliency instead of LLM logic atoms.

J. Robust Fake News Detection against LLM-Empowered Style Attacks

1) Title, Authors, Date and Publisher

"Fake news in sheep's clothing: Robust fake news detection against LLM-empowered style attacks", by J. Wu, J. Guo and B. Hooi; published August 2024 in Proc. 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2024), ACM [27].

2) Problem and Objectives

An LLM can rewrite a fake article in the voice of a trusted outlet. The paper asks how badly such style attacks hurt text-based detectors and how to make detectors robust to them.

3) Methodology Adopted

The authors show that LLM-camouflaged fake news reduces the F1 score of state-of-the-art detectors by up to 38%. Their detector, SheepDog, is trained on LLM-generated reframings of each article in different styles with a consistency objective, and uses content-focused veracity rationales from an LLM as extra weak supervision, giving style-agnostic predictions.

4) Performance and Limitations

SheepDog improves robustness on three benchmarks, but it relies heavily on LLMs during training and is evaluated on English only. Its central finding is a warning for any system that depends on writing style: stylistic cues can be forged, so they must be combined with content evidence.

5) Relevance to TCF-Net

Warns against relying on style alone; in TCF-Net stylometry is one gated stream that semantics can override.

K. ASPEST: Active Selective Prediction under Distribution Shift

1) Title, Authors, Date and Publisher

"ASPEST: Bridging the gap between active learning and selective prediction", by J. Chen, J. Yoon, S. Ebrahimi, S. Ö. Arik, S. Jha and T. Pfister; published March 2024 in Transactions on Machine Learning Research (TMLR) [28].

2) Problem and Objectives

Selective prediction lets a model abstain on uncertain inputs, but under distribution shift the model becomes over-confident and abstains on the wrong cases, while collecting human labels is costly. The paper aims to improve accuracy and coverage under shift with a small labelling budget.

3) Methodology Adopted

The authors define active selective prediction, in which the model queries human labels for the most informative uncertain samples from the shifted domain. Their method, ASPEST, uses checkpoint ensembling and self-training to reduce over-confidence and outperforms prior active-learning and selective-prediction baselines on several datasets.

4) Performance and Limitations

ASPEST is a general-purpose method and has not been evaluated on misinformation or Indian-language text. It shows, however, how an abstaining system can keep improving by learning from the cases it defers, a natural extension of the human-in-the-loop workflow of TCF-Net.

5) Relevance to TCF-Net

Provides the route for extending TCF-Net abstention to distribution shift by learning from deferred claims.

L. Thematic Synthesis

Read together, the eleven works show four clear trends. First, multilingual transformers have become the default backbone for Indian-language misinformation: Hindi systems use XLM-RoBERTa, mBERT and ELECTRA [19], Dravidian systems use MuRIL [20], and the Malayalam shared task was dominated by fine-tuned XLM-RoBERTa and mBERT [25]. These models work well on native-script news, but the reviewed evaluations rarely include short, Romanised, code-mixed posts.

Second, there is a steady move towards richer evidence. Linguistic features improve Hindi detection when combined with embeddings [24]; graph methods add propagation structure [6]; multimodal methods add images [22]; and LLM-based methods add world knowledge and rationales [23], [26]. Each source helps, but each also brings a cost: graphs need platform data, images need multimodal posts, and LLMs need expensive queries. The cheapest sources, writing style and a text-derived spread prior, are available for every post at no extra cost, which motivates their use in TCF-Net.

Third, robustness and trust have become explicit goals. TELLER makes decisions inspectable through logic rules [26], and SheepDog shows that detectors which rely on style alone can be fooled by LLM rewrites [27]. The second finding is directly relevant here: it argues for using stylometry as one gated stream that semantics can override, rather than as the sole signal. Fourth, abstention is still treated as a separate machine-learning topic [28]; none of the misinformation systems reviewed can say "I do not know", and none studies how many unverifiable claims a deployed system should defer to people.

Finally, hate speech and misinformation remain separate research lines. The code-mixed hate work [21] and the fake-news work share datasets, languages and models, yet no reviewed system addresses both. TCF-Net sits at the intersection of these trends: a multilingual transformer backbone, cheap complementary evidence, a gate that limits over-reliance on style, an explicit reject option and a joint hate-speech head.

M. Comparative Summary

Table I summarises the four review criteria for all eleven works, and Table II compares them with TCF-Net on the properties that matter most here. Only ASPEST offers a reject option and only Sreelakshmi et al. address hate speech; no work combines veracity, hate speech and abstention, and none targets Romanised code-mixed posts.

TABLE I
LITERATURE REVIEW SUMMARY (11 WORKS, 2023–2024)

Work	(1) Publication	(2) Problem	(3) Methodology	(4) Limitations
A. Praseed *et al.* [19]	Engineering Applications of Artificial Intelligence, 2023	Hindi is one of the most widely spoken languages in the world, yet labelled resources and models for detecting fake Hindi news lag far behind those for English.	Several pre-trained multilingual transformers, namely XLM-RoBERTa, multilingual BERT and ELECTRA, are fine-tuned independently on Hindi news.	The study covers Devanagari Hindi only, so Romanised Hinglish posts, which dominate informal platforms, are not addressed.
E. Raja *et al.* [20]	20th International Conference on Natural Language Processing (ICON 2023), December 2023	Tamil, Telugu, Kannada and Malayalam have little annotated data for fake-news detection, and fine-tuning large models on them is limited by GPU memory.	The MuRIL transformer, pre-trained on Indian languages, is fine-tuned with gradient accumulation, which simulates large batches inside a small memory budget, together with adaptive learning-rate scheduling.	It relies on semantics alone and ignores stylistic and propagation cues.
K. Sreelakshmi *et al.* [21]	IEEE Access, 2024	Hate and offensive speech in Kannada–English, Malayalam–English and Tamil–English code-mixed posts is hard to detect because the text is noisy and transliterated, and the classes are heavily imbalanced, with far fewer offensive than benign posts.	Posts are represented with multilingual transformer embeddings that are classified by classical machine learning models.	The embeddings are used as fixed features rather than fine-tuned, and the system addresses offensive language only: a separate model would be needed to judge veracity.
H. T. Phan *et al.* [6]	Applied Soft Computing, May 2023	Fake news is shaped not only by its content but by who shares it and how it spreads.	The authors review the state of the art, outline a general procedure for building a GNN-based detector, and propose a taxonomy that groups models by the family of graph neural network they use.	It also exposes a practical weakness: GNN methods need sharing graphs and user data that are often private, unavailable or not yet formed when a post first appears, and most benchmarks are English..
S. Tufchi *et al.* [22]	International Journal of Multimedia Information Retrieval, 2023	Modern misinformation combines text with images and other media, and single-modality detectors miss inconsistencies between them.	The authors systematically review multimodal detectors, compare classification models across text- and image-based fake-news datasets, and discuss feature extraction and fusion strategies such as early, late and attention-based fusion before outlining open challenges..	Its focus, however, is image–text fusion on largely English and Chinese benchmarks; it gives little attention to code-mixed Indian languages, to stylometric evidence as a modality, or to selective prediction..
B. Hu *et al.* [23]	Proceedings of the AAAI Conference on Artificial Intelligence, March 2024	Large language models (LLMs) such as GPT-3.5 have broad world knowledge, but it was unclear whether they detect fake news better than smaller fine-tuned models.	An empirical study first compares GPT-3.5 with a fine-tuned BERT detector.	Querying an LLM adds cost and latency, the evaluation covers English and Chinese only, and the system still forces a decision on every item..
R. Sharma *et al.* [24]	ACM Transactions on Asian and Low-Resource Language Information Processing, June 2023	Hindi fake-news research lacked a sizeable annotated corpus and representations that capture the linguistic character of deceptive Hindi text.	The authors introduce the HinFakeNews dataset of roughly 33,300 annotated Hindi articles and a linguistic-feature-based word embedding (LFWE) that enriches word representations with hand-crafted linguistic features before classification.	The work treats veracity in isolation, with no hate-speech screening and no mechanism to abstain on claims that cannot be verified.
M. Subramanian et al. *et al.* [25]	3rd Workshop on Speech and Language Technologies for Dravidian Languages (DravidianLangTech), September 2023	Fake-news detection for Dravidian social media lacked a shared benchmark.	The organisers collected 5,091 posts and comments from YouTube, split them into training, validation and test sets, and evaluated systems with macro-F1.	The benchmark covers a single language and all submitted systems are forced classifiers without abstention or explanation..
H. Liu *et al.* [26]	Findings of the Association for Computational Linguistics: ACL 2024, August 2024	Neural fake-news detectors are opaque, generalise poorly to new domains and are hard to correct.	TELLER uses a dual-system design.	It explains decisions but still outputs a verdict for every claim, and it does not address code-mixed text or hate speech..
J. Wu *et al.* [27]	30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2024), August 2024	An LLM can rewrite a fake article in the voice of a trusted outlet.	The authors show that LLM-camouflaged fake news reduces the F1 score of state-of-the-art detectors by up to 38%.	Its central finding is a warning for any system that depends on writing style: stylistic cues can be forged, so they must be combined with content evidence..
J. Chen *et al.* [28]	Transactions on Machine Learning Research (TMLR), March 2024	Selective prediction lets a model abstain on uncertain inputs, but under distribution shift the model becomes over-confident and abstains on the wrong cases,	The authors define active selective prediction, in which the model queries human labels for the most informative uncertain samples from the shifted domain.	It shows, however, how an abstaining system can keep improving by learning from the cases it defers, a natural extension of the human-in-the-loop workflow

Work	(1) Publication	(2) Problem	(3) Methodology	(4) Limitations
		while collecting human labels is costly.		of TCF-Net..

TABLE II
COMPARISON OF REVIEWED WORKS WITH TCF-NET

Work	Task / language	Core technique	Abst.	Hate
A. Praseed *et al.* [19]	Fake news, Hindi	Transformer ensemble (XLM-R, mBERT, ELECTRA)	No	No
E. Raja *et al.* [20]	Fake news, Dravidian	MuRIL + gradient accumulation	No	No
K. Sreelakshmi *et al.* [21]	Hate speech, Dravidian code-mixed	Transformer embeddings + cost-sensitive ML	No	Yes
H. T. Phan *et al.* [6]	Fake news (survey)	Taxonomy of GNN detectors	–	–
S. Tufchi *et al.* [22]	Fake news (survey), multimodal	Review of text–image fusion	–	–
B. Hu *et al.* [23]	Fake news, English / Chinese	LLM rationales distilled into a small model	No	No
R. Sharma *et al.* [24]	Fake news, Hindi	Linguistic-feature word embeddings	No	No
M. Subramanian et al. *et al.* [25]	Fake news, Malayalam social media	Shared task (XLM-R, mBERT systems)	No	No
H. Liu *et al.* [26]	Fake news, English	LLM logic atoms + learned rules	No	No
J. Wu *et al.* [27]	Fake news, English	Style-agnostic training with LLM reframings	No	No
J. Chen *et al.* [28]	General classification	Active selective prediction	Yes	No
TCF-Net (this work)	**Fake news + hate, 5 languages incl. code-mixed**	**MuRIL + stylometry + SIR, gated fusion**	**Yes**	**Yes**

Table III lists the datasets used or released by the reviewed works and by this project, which shows how few resources cover short, Romanised, code-mixed posts with both veracity and hate labels.

TABLE III
DATASETS IN THE REVIEWED LITERATURE AND THIS WORK

Dataset	Language(s)	Size	Labels
DFND [20]	Tamil, Telugu, Kannada, Malayalam	27,000+ articles	Fake / real
HinFakeNews [24]	Hindi	≈33,300 articles	Fake / real
DravidianLangTech 2023 [25]	Malayalam (social media)	5,091 posts	Fake / original
HASOC 2021 [7]	English, Hindi, Marathi	–	Hate / offensive
DravidianCodeMix [8]	Tamil, Malayalam, Kannada code-mixed	–	Sentiment, offensive
LIAR [4]	English	12,836 statements	Six-way truthfulness
TCF-Net benchmark (ours)	**EN, TA, Tanglish, Hinglish, HI**	**5,000 claims**	**Veracity, hate, ambiguity**

N. Research Gaps

- **Single-stream evidence.** Most detectors read only the semantics of a text. Stylistic deception cues and propagation behaviour are rarely fused in one model, and GNN approaches need sharing graphs that are often unavailable at posting time.
- **Code-mixed gap.** Indian-language detectors are built on native-script news; Romanised Tanglish and Hinglish posts, a major channel of online news, are seldom addressed.
- **No reject option.** None of the reviewed detectors can decline a claim that cannot be verified from the text, which is risky for fact-checking workflows.
- **Isolated treatment.** Misinformation and hate speech often appear together but are studied with separate models.
- **Optimistic evaluation.** Templated corpora with random splits can inflate accuracy, and grouped evaluation is seldom reported.
- **Forgeable style.** LLM rewrites in a trusted style defeat style-reliant detectors [27], so stylometry should be one gated stream among several rather than the only one.

IV. PROPOSED METHODOLOGY AND SYSTEM DESIGN

A. Design Principles

TCF-Net follows four principles derived from the research gaps: (1) evidence diversity, combining what a claim says, how it is written and how it would spread; (2) language robustness, using representations that tolerate script mixing and spelling variation; (3) calibrated humility, answering only when confident and deferring the rest; and (4) transparency, exposing per-sample evidence weights and word-level attributions.

B. System Architecture

Fig. 1 shows the architecture. The input layer accepts a claim in any of the five language settings. The evidence layer computes the semantic, stylometric and propagation representations. The projection layer maps each to a 256-dimensional vector through Linear–LayerNorm–GELU–Dropout blocks. The fusion layer computes per-sample gate weights and a weighted sum, followed by a shared trunk. The output layer contains the veracity head f(x), the selection head g(x), the auxiliary head h(x) and the hate-speech head, plus a decision rule that converts them into REAL, FAKE or NOT SURE.

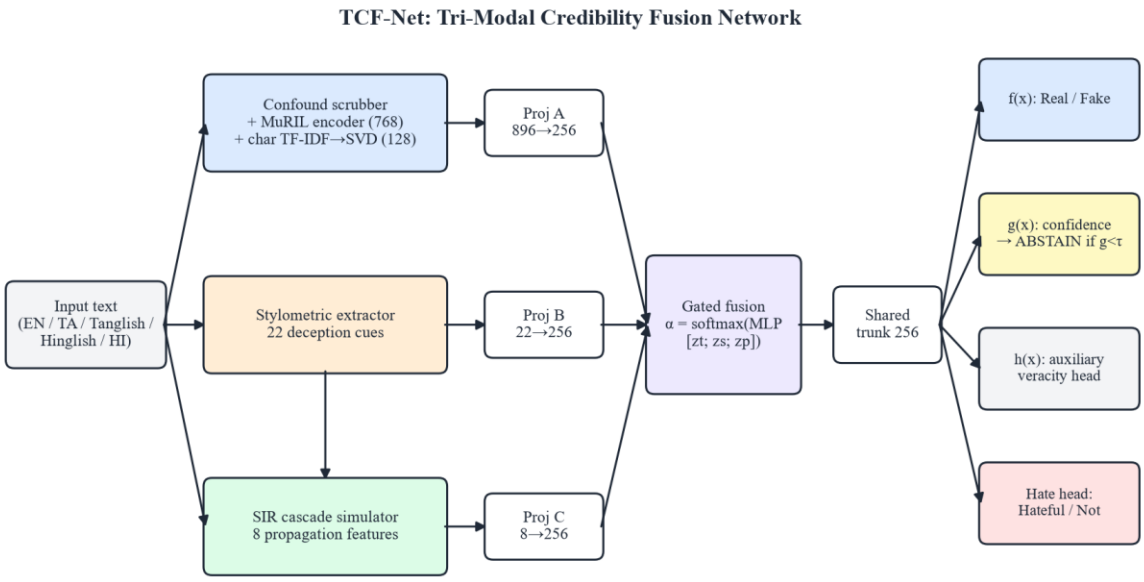


Fig. 1. System architecture of TCF-Net.

C. Processing Pipeline

Fig. 2 traces a claim from input to decision: pre-processing, tri-modal feature extraction, standardisation with statistics fitted on the training split, gated fusion and the trunk, the four heads, and finally the decision rule. Every transformation that has parameters (TF-IDF vocabulary, SVD basis, scalers and the abstention threshold) is fitted without access to test data.

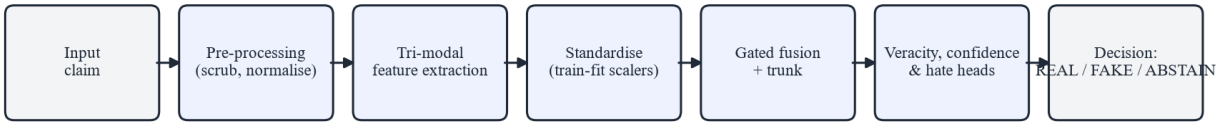


Fig. 2. End-to-end processing pipeline from input claim to decision.

D. Confound Scrubbing

Before semantic encoding, the text passes through a scrubber that applies Unicode NFKC normalisation, maps typographic quotes and dashes to ASCII, removes agency datelines such as "WASHINGTON (Reuters) -", strips URLs, social-media handles and tags such as "(VIDEO)", standardises contractions and lower-cases the result. These artefacts identify who published a story rather than what it claims, and a classifier that learns them would fail on new sources. The stylometric stream deliberately receives the raw text, because capitals and exclamation marks are genuine signals of sensational writing.

E. Modality A: Contextual Semantics

The scrubbed claim is encoded by MuRIL [12] (maximum 128 tokens) and mean-pooled into $e \in \mathbb{R}^{768}$. In parallel, a TF-IDF vectoriser over character 2–5-grams (word-boundary aware, up to 30,000 features, fitted on the training split) is compressed by truncated SVD to $l \in \mathbb{R}^{128}$. The semantic input is the concatenation

$$x_t = standardise([e\ ;\ l]) \in \mathbb{R}^{896} \quad (2)$$

Character n-grams complement the encoder on Romanised spellings and on Indic combining marks, and they are cheap to compute.

F. Modality B: Deception Stylometry

Twenty-two features are computed from the raw text (Table IV). They cover length and sentence structure, punctuation and casing, lexicon densities, pronoun use, lexical diversity, readability and affect. Lexicons contain entries for English, Hindi and Hinglish, and Tamil and Tanglish, for example "shayad" (perhaps), "khulasa" (exposé), "oruvela" (maybe), அதிர்ச்சி (shock) and சதி (conspiracy). A script-aware tokenizer treats Devanagari and Tamil code points, including vowel signs and viramas, as word characters so that native-script words remain intact and match their lexicon entries. Densities are normalised by word count.

TABLE IV
THE 22 STYLOMETRIC DECEPTION FEATURES

#	Feature	Description	#	Feature	Description
1	char_len	Text length in characters	12	hedge_density	Hedging words (allegedly, shayad, oruvela)
2	word_len	Number of words	13	sensational_density	Sensational words (shocking, khulasa, adhirchi)
3	avg_word_len	Mean word length	14	superlative_density	Superlatives (biggest, sabse)
4	sent_count	Number of sentences	15	negation_density	Negations (not, nahi, illa)
5	avg_sent_len	Mean words per sentence	16	first_person	First-person pronouns
6	excl_ratio	Exclamation marks per character	17	second_person	Second-person pronouns
7	ques_ratio	Question marks per character	18	modal_density	Modal verbs (will, must, might)
8	caps_word_ratio	Share of ALL-CAPS words	19	lexical_diversity	Type–token ratio
9	digit_ratio	Share of digits	20	flesch_ease	Flesch reading ease (scaled)
10	punct_ratio	Share of punctuation	21	affect_polarity	Positive minus negative affect words
11	quote_ratio	Quotation density	22	punct_burstiness	Clustering of punctuation runs

The lexicon features are densities $d_L = |\{w \in x : w \in L\}| / |x|$ over the tokens of the claim. Punctuation burstiness is the standard deviation of the gaps between consecutive punctuation marks, which is high when marks are clustered as in "!!!" and low when they are spread evenly. Lexical diversity is the type–token ratio, and affect polarity is the difference between positive and negative affect densities. Table V summarises the lexicons with examples from each language group.

TABLE V
MULTILINGUAL LEXICONS USED BY THE STYLOMETRIC STREAM

Lexicon	Size	English examples	Hindi / Hinglish examples	Tamil / Tanglish examples
Hedges	32	allegedly, reportedly, unconfirmed	shayad, afwaah, शायद	oruvela, கூறப்படுகிறது, வதந்தி
Sensational	35	shocking, exposed, bombshell	khatarnak, khulasa, सनसनीखेज	adhirchi, அதிர்ச்சி, சதி, பரபரப்பு
Superlatives	20	biggest, ultimate, never	sabse, ekdam, सबसे	periya, மிகப்பெரிய, முழுவதும்
Negations	25	not, nobody, nothing	nahi, jhooth, नहीं	illa, vendam, இல்லை, பொய்
First person	18	I, we, our	main, hum, hamara	naan, namba, நாம்
Second person	14	you, your	aap, tum, tera	neenga, ungala, நீங்கள்
Positive affect	18	proven, trusted, success	shandar, badhiya, शानदार	nalladhu, வெற்றி, நன்மை
Negative affect	26	fraud, panic, threat	dhokha, barbaad, खतरा	thurogi, mosadi, துரோகம்

G. Modality C: SIR Propagation Prior

Emotionally arousing content is shared more readily. TCF-Net turns this into features without platform data. An arousal score is computed from three stylometric cues and sets the rates of a discrete-time SIR model [14], [15]:

$$a = clip(4 \cdot excl + 3 \cdot caps + 6 \cdot sens,\ 0,\ 1) \quad (3)$$

$$\beta = 0.18 + 0.45a + \varepsilon_1, \quad \gamma = 0.22 - 0.08a + \varepsilon_2 \quad (4)$$

$$S_{k+1} = S_k - \beta S_k I_k \quad (5)$$

$$I_{k+1} = I_k + \beta S_k I_k - \gamma I_k, \quad R_{k+1} = R_k + \gamma I_k \quad (6)$$

where ε_1 and ε_2 are small Gaussian perturbations with a fixed seed. The cascade runs for 24 steps from $I_0 = 0.002$. Eight descriptors form x_p: peak infection, normalised time to peak, virality slope, cascade breadth, depth, an echo-chamber score, a bot-likeness score and the decay rate after the peak. The simulator never sees the label, so it cannot leak ground truth. Fig. 3 contrasts a sensational and a neutral claim.

TABLE VI
THE EIGHT SIR PROPAGATION DESCRIPTORS

Descriptor	Definition
peak_infected	max over k of I_k
time_to_peak	argmax I_k / 24
virality_slope	peak / time-to-peak (steps)
cascade_breadth	$R_{24} + I_{24}$, share ever reached
cascade_depth	log(1 + 24·peak)
echo_chamber	clip(0.3 + 0.5a + noise, 0, 1)
bot_likeness	clip(0.1 + 0.6a·u, 0, 1), u ~ U(0.5, 1.2)
decay_rate	(peak − min after peak) / peak

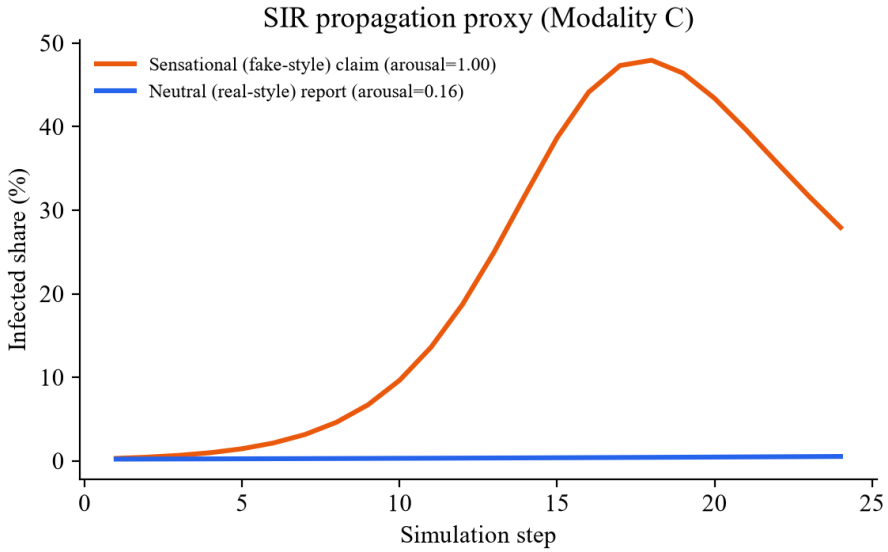


Fig. 3. SIR propagation prior for a sensational versus a neutral claim.

H. Cross-Modal Gated Fusion

Each modality is projected to z_t, z_s, z_p ∈ $\mathbb{R}^{256}$. A two-layer gate looks at all three and outputs a softmax distribution over modalities that weights their sum:

$$\alpha = softmax(W_2\, ReLU(W_1[z_t\,;\, z_s\,;\, z_p])) \quad (7)$$

$$z = \alpha_1 z_t + \alpha_2 z_s + \alpha_3 z_p, \quad u = Trunk(z) \quad (8)$$

Because α is computed for every sample, the model can lean on stylometry for loud clickbait and on semantics for calm but false statements. An entropy bonus $H(\alpha) = -\Sigma\, \alpha_m \log \alpha_m$ in the loss discourages early collapse onto a single stream. The trunk is a Linear–LayerNorm–GELU–Dropout block of width 256.

I. Prediction Heads

Four heads read the trunk output u: the veracity head f(u) ∈ $\mathbb{R}^2$ gives logits for Real and Fake; the selection head g(u) ∈ [0, 1] is a two-layer network with a sigmoid output that estimates whether the prediction should be trusted; the auxiliary head h(u) ∈ $\mathbb{R}^2$ is a second veracity classifier trained on all samples; and the hate head f_hate(u) ∈ $\mathbb{R}^2$, with its own auxiliary head, predicts Hateful versus Not Hateful.

J. Training Objective

Following SelectiveNet [17], with batch coverage $\varphi = (1/B)\, \Sigma\, g(x_i)$ and target coverage c = 0.85, the selective loss is

$$\hat{r} = \Sigma_i\, g(x_i) \cdot CE(f(x_i), y_i) \,/\, \Sigma_i\, g(x_i) \quad (9)$$

$$L_sel = \hat{r} + \lambda \cdot \max(0,\, c - \varphi)^2, \quad \lambda = 32 \quad (10)$$

$$L_ver = 0.5 \cdot L_sel + 0.5 \cdot CE(h(x), y) \quad (11)$$

$$L_hate = 0.7 \cdot CE(f_hate(x), y_h) + 0.3 \cdot CE(h_hate(x), y_h) \quad (12)$$

$$L = L_ver - 0.02 \cdot H(\alpha) + 0.5 \cdot L_hate \quad (13)$$

The auxiliary loss keeps the trunk learning from samples the selection head down-weights. Samples without a hate label are masked out of L_hate, so corpora annotated only for veracity can still be used [18].

K. Risk-Targeted Abstention Calibration

At inference the model answers only when g(x) ≥ τ. A threshold set as a fixed coverage quantile drifts when the share of ambiguous claims differs between splits, so τ is instead calibrated on the validation split to control risk:

1. Predict P(fake | x) and g(x) for every validation claim.
2. Sort the claims by g(x) in descending order.
3. Compute the cumulative selective error of the top-k claims for every k.
4. Choose the largest k whose selective error is at most 1%.
5. Set τ midway between the k-th and (k+1)-th scores so it lies inside the confidence gap.

For the trained model this gives τ = 0.375. The test set is never used for calibration.

L. Inference Workflow

At run time a claim is scrubbed, encoded by MuRIL and the character-n-gram projector, described by the 22 stylometric features and the 8 propagation descriptors, standardised with the saved scalers and passed through the network. The decision is NOT SURE if g(x) < τ; otherwise FAKE if P(fake | x) ≥ 0.5 and REAL if not. The hate verdict is reported independently. For explanation, the application shows α and recomputes P(fake) with each word removed in turn to obtain occlusion saliency.

The complete procedures are summarised as Algorithm 1 (training) and Algorithm 2 (inference).

1) Algorithm 1 – Training

1. Input: labelled claims (x, y, y_h); backbone MuRIL; seed 42.
2. Scrub every claim and encode it with MuRIL; cache the 768-D embeddings.
3. Compute the 22 stylometric features from raw text and the 8 SIR descriptors from them.
4. Group claims by scrubbed text and split with StratifiedGroupKFold into train, validation and test.

5. Fit character TF-IDF, truncated SVD and the three standard scalers on the training split only.
6. For each epoch: shuffle mini-batches of 64; compute L; clip gradients at 1.0; take an AdamW step; anneal the learning rate.
7. After each epoch evaluate validation accuracy and ROC-AUC; keep the best state; stop after 10 epochs without improvement.
8. Calibrate τ on the validation split with risk-targeted calibration.
9. Output: network weights, scalers, projector, τ and evaluation artifacts.

2) Algorithm 2 – Inference

1. Input: a claim x in any supported language.
2. Compute x_t from the scrubbed text (MuRIL + character n-grams), x_s from the raw text and x_p from x_s.
3. Standardise with the saved scalers and run the network to obtain P(fake), g(x), α and P(hate).
4. If $g(x) < \tau$ return NOT SURE; otherwise return FAKE if $P(\text{fake}) \geq 0.5$, else REAL.
5. Return HATEFUL if $P(\text{hate}) \geq 0.5$, together with the gate weights α and word-level occlusion saliency.

M. Model Complexity

The MuRIL encoder (237,556,224 parameters) is frozen and used only as a feature extractor. The trainable network has 538,380 parameters, distributed as in Table VII; the semantic projection and the gate dominate because they read the 896- and 768-dimensional inputs.

TABLE VII
TRAINABLE PARAMETERS OF TCF-NET

Component	Parameters
Projection A (semantic)	230,144
Projection B (stylometric)	6,400
Projection C (propagation)	2,816
Gated fusion	197,635
Shared trunk	66,304
Veracity head f(x)	514
Selection head g(x)	33,025
Auxiliary head h(x)	514
Hate heads	1,028
Total (trainable)	**538,380**

N. Comparison with Existing Approaches

TABLE VIII
EXISTING APPROACHES VERSUS THE PROPOSED SYSTEM

Aspect	Typical existing systems	Proposed TCF-Net
Evidence	Text semantics only	Semantics + stylometry + propagation prior
Languages	English or one Indic language	English, Tamil, Tanglish, Hinglish, Hindi
Output	Forced binary label	Real / Fake / Not Sure with g(x)
Explainability	Post-hoc, if any	Gate weights, stylometric profile, word saliency
Hate speech	Separate model	Joint head on the shared trunk
Evaluation	Random split	Grouped split + risk–coverage analysis

V. IMPLEMENTATION

A. Development Environment

TCF-Net is implemented in Python with PyTorch [29], Hugging Face Transformers [30] and scikit-learn [31]. Training ran on a CPU-only Windows 11 laptop; the code also supports CUDA with automatic mixed precision. Table IX lists the environment.

TABLE IX
DEVELOPMENT ENVIRONMENT

Component	Specification
Language	Python 3.12.10
Deep learning	PyTorch 2.14.0
NLP library	Transformers 4.57.6
Classical ML	scikit-learn 1.9.0
Text features	textstat, custom multilingual lexicons
Encoder	google/muril-base-cased (frozen)
Interface	Gradio 3-tab app, Plotly charts
Report figures	Matplotlib
Hardware	CPU (CUDA + AMP supported)
OS	Windows 11, 64-bit

B. Benchmark Construction

The benchmark was generated with data/generate_dataset.py from language-specific template pools. For each sample the generator draws a language with fixed proportions (English 35%, Tamil 20%, Tanglish 20%, Hinglish 15%, Hindi 10%) and a claim type. Sensational deceptions combine a sensational opener, a topic and a fabricated consequence, followed by an urge to share; a fraction of them receive an attached hateful phrase in the same language. Clear facts combine an official organisation, a topic and a factual, measured verb phrase. Borderline claims are unconfirmed rumours ("officials may consider…, though formal documentation is still awaited") whose veracity label is assigned at random. Hindi contains only clear and sensational claims.

C. Dataset Statistics

Table X and Fig. 4 summarise the 5,000 claims. Veracity is nearly balanced (2,530 fake, 2,470 real), 597 claims are hateful, and 788 (15.8%) are borderline. Because labels of borderline claims are random, no model can exceed roughly 92% accuracy when forced to answer everything; these claims exist to test whether the system abstains.

TABLE X
COMPOSITION OF THE FIVE-LANGUAGE BENCHMARK

Language	Samples	Real	Fake	Hateful	Border.
English	1,754	892	862	202	298
Hindi	516	277	239	0	0
Hinglish	746	345	401	112	138
Tamil	949	448	501	122	171
Tanglish	1,035	508	527	161	181
Total	**5,000**	**2,470**	**2,530**	**597**	**788**

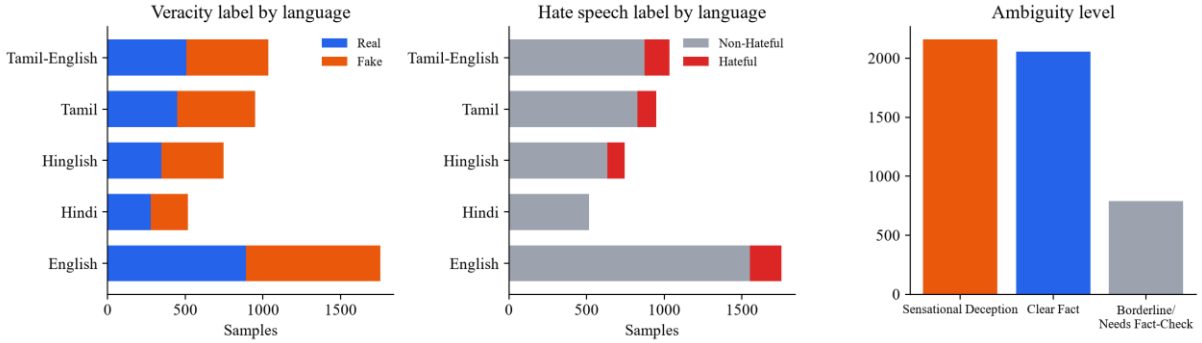


Fig. 4. Label distribution by language and ambiguity level.

D. Representative Samples

Table XI shows one fake and one real claim for each language, illustrating the scripts and the stylistic contrast between sensational and official wording.

TABLE XI
REPRESENTATIVE CLAIMS FROM EACH LANGUAGE

Language	Statement (truncated)	Veracity	Hate
Hinglish	KHATARNAK SACH EXPOSED!! Dekhiye ye viral sach!! Vaccination Drive aaj raat se illegal declare hone wala hai. Jaldi se jaldi share karein sabhi dosto …	Fake	Hateful
Hinglish	Supreme Court Public Relations Office ne 500 rupee notes ke liye new guidelines official portal pe upload kar diya hai.	Real	Non-Hateful
English	MUST SHARE BEFORE DELETED: Clean Energy Policy poisons local groundwater. Share before authorities delete this link!!!	Fake	Non-Hateful
English	The Peer-reviewed Medical Journal has issued official regulatory guidance on clean energy policy during its scheduled review on 14 March.	Real	Non-Hateful
Tamil	அதிர்ச்சி செய்தி!! ரகசிய உண்மை வெளியானது!! மழை வெள்ள முன்னெச்சரிக்கை மக்களை ஏமாற்றும் மிகப்பெரிய சதி! உடனடியாக வாட்ஸ்அப்பில் அனைவருக்கும் பகிருங்கள்.	Fake	Non-Hateful
Tamil	தமிழக சுகாதாரத்துறை புதிய தடுப்பூசி தொடர்பாக அதிகாரப்பூர்வ செய்திக்குறிப்பு இன்று வெளியிடப்பட்டுள்ளது.	Real	Non-Hateful
Tanglish	BAYANGARAMAANA UNMAI: Doctors hide panraanga!! Electricity Tariff Revision secret chip insert panraanga nu doctor reveal pannitaaru! Please share with…	Fake	Non-Hateful
Tanglish	TNPSC Official Board has officially announced that bus fare subsidy official website-la formal notification update pannirukaanga.	Real	Non-Hateful
Hindi	वायरल अलर्ट: स्वास्थ्य विभाग ने छिपाई यह गंभीर बात: राष्ट्रीय स्वास्थ्य मिशन को लेकर अधिकारियों पर लगे बड़े आरोप। इस संदेश को अधिक से अधिक साझा करें।	Fake	Non-Hateful
Hindi	भारतीय रिजर्व बैंक (RBI) ने आज एक आधिकारिक प्रेस विज्ञप्ति जारी कर नई शिक्षा नीति की प्रगति रिपोर्ट प्रस्तुत की है।	Real	Non-Hateful

E. Leak-Free Grouped Split

The corpus contains only 1,531 distinct texts. Samples are grouped by scrubbed text and split with StratifiedGroupKFold (stratified by veracity and language): one of five folds is the test set, and one sixth of the remainder is the validation set (Table XII). No text appears in more than one split.

TABLE XII
DATA SPLIT PRODUCED BY THE GROUPED PROTOCOL

Split	Samples	Share	Used for
Train	3,371	67.4%	Weights, scalers, TF-IDF/SVD
Validation	630	12.6%	Early stopping, τ calibration
Test	999	20.0%	Final evaluation only

F. Software Architecture and Modules

The code base is organised into data, features, models and evaluation packages, a training script and two user interfaces (Table XIII).

TABLE XIII
SOURCE MODULES AND RESPONSIBILITIES

Module	Responsibility
data/generate_dataset.py	Builds the five-language benchmark
data/loaders.py	Benchmark, ISOT, LIAR, code-mixed loaders
features/scrubber.py	Removes publisher confounds
features/stylometric.py	22-D multilingual stylometry
features/diffusion.py	8-D SIR cascade simulator
models/tcf_net.py	Network, gated fusion, selective loss
train.py	Split, training, evaluation, baselines, ablation
evaluation/metrics.py	Metrics, risk–coverage, AURC
evaluation/make_figures.py	Report figures from artifacts
evaluate_cross_dataset.py	Out-of-domain audit
app.py / streamlit_app.py	Interactive web applications

G. Training Procedure

Training uses AdamW [32] with cosine learning-rate annealing, gradient clipping at norm 1.0 and early stopping on validation accuracy (ties broken by ROC-AUC) with patience 10. The procedure is: (1) load and scrub the corpus; (2) extract and cache MuRIL embeddings; (3) extract stylometric and SIR features; (4) build the grouped split; (5)

fit TF-IDF, SVD and scalers on the training split; (6) train the network; (7) calibrate τ on validation data; (8) evaluate on the test split, train baselines and ablations on the same split, and export all artifacts. Table XIV lists the hyper-parameters; training stopped after 13 epochs.

TABLE XIV
TRAINING HYPER-PARAMETERS

Hyper-parameter	Value
Fusion width / dropout	256 / 0.3
Optimiser	AdamW, lr 0.001, weight decay 1e-4
Batch size	64
Max epochs / patience	40 / 10
Epochs run	13
Target coverage c	0.85
Coverage penalty λ	32
Gate entropy weight	0.02
Hate loss weight	0.5
Target validation risk	0.01
Calibrated τ	0.375
Random seed	42

H. Web Application

The application (app.py) is built with Gradio and Plotly and has three tabs. Tab 1, the Live Credibility and Hate Speech Inspector, accepts a claim and a coverage slider and shows the veracity verdict, the hate verdict, a gauge for g(x), a donut chart of the modality weights α and a bar chart of word-level occlusion saliency. Tab 2 shows the stylometric profile and the simulated SIR infection curve of a claim. Tab 3 plots the risk–coverage curve against random abstention and shows abstention-audit tables read from the saved test results. A Streamlit variant adds a dataset explorer.

TABLE XV
USER-INTERFACE COMPONENTS OF THE WEB APPLICATION

Tab	Component	Purpose
1	Claim box, coverage slider	Input and desired operating point
1	Veracity and hate badges	REAL / FAKE / NOT SURE and HATEFUL verdicts
1	Confidence gauge	g(x) against the threshold
1	Modality donut	Per-claim gate weights α
1	Occlusion saliency bars	Words pushing towards FAKE or REAL
2	Stylometric chart	Values of key deception cues
2	SIR curve	Simulated infected share over 24 steps
3	Risk–coverage plot, slider	Error versus coverage, random baseline
3	Audit tables	Accuracy and abstention by claim type and language

Training writes its outputs to the artifacts folder (Table XVI); the web application and every figure and table in this report are generated from these files, so the report always reflects the deployed model.

TABLE XVI
ARTIFACTS PRODUCED BY THE TRAINING PIPELINE

File	Content
tcfnet.pt	Network weights, dimensions, configuration, τ
scalers.joblib	Scalers, TF-IDF, SVD, backbone name
final_metrics.json	Veracity, selective and hate metrics
risk_coverage_curve.csv	Selective and random risk per coverage
per_language_metrics.csv	Accuracy, coverage, hate accuracy by language
per_ambiguity_metrics.csv	The same by claim type
baseline_comparison.csv	Baselines and ablations
test_predictions.csv	Per-claim P(fake), g(x), α, P(hate)
training_history.csv	Loss and validation metrics per epoch
run_info.json	Dataset, split sizes, epochs

I. Running the System

After installing the dependencies with "pip install -r requirements.txt", the demonstration starts with "python app.py" and opens at http://localhost:7860. Retraining from scratch runs with "python train.py --backbone google/muril-base-cased", which takes about five minutes on a CPU once the encoder is cached, and "python -m evaluation.make_figures" regenerates every figure in this report.

VI. SYSTEM DEMONSTRATION

This section walks through the complete flow from input to output in the web application, on native Tamil text and on code-mixed and English claims. Every screenshot was captured from the running application with the trained model.

A. Tamil Fake and Hateful Claim

Fig. 5 shows a sensational Tamil claim that a new ration-card rule is a conspiracy, followed by a call to drive a "traitor gang" out of Tamil Nadu. TCF-Net labels it FAKE with confidence 1.000 and HATEFUL. The gate spreads weight across all three modalities, and occlusion saliency highlights the inflammatory words துரோகி ("traitor"), விரட்ட ("drive out") and சதி ("conspiracy").

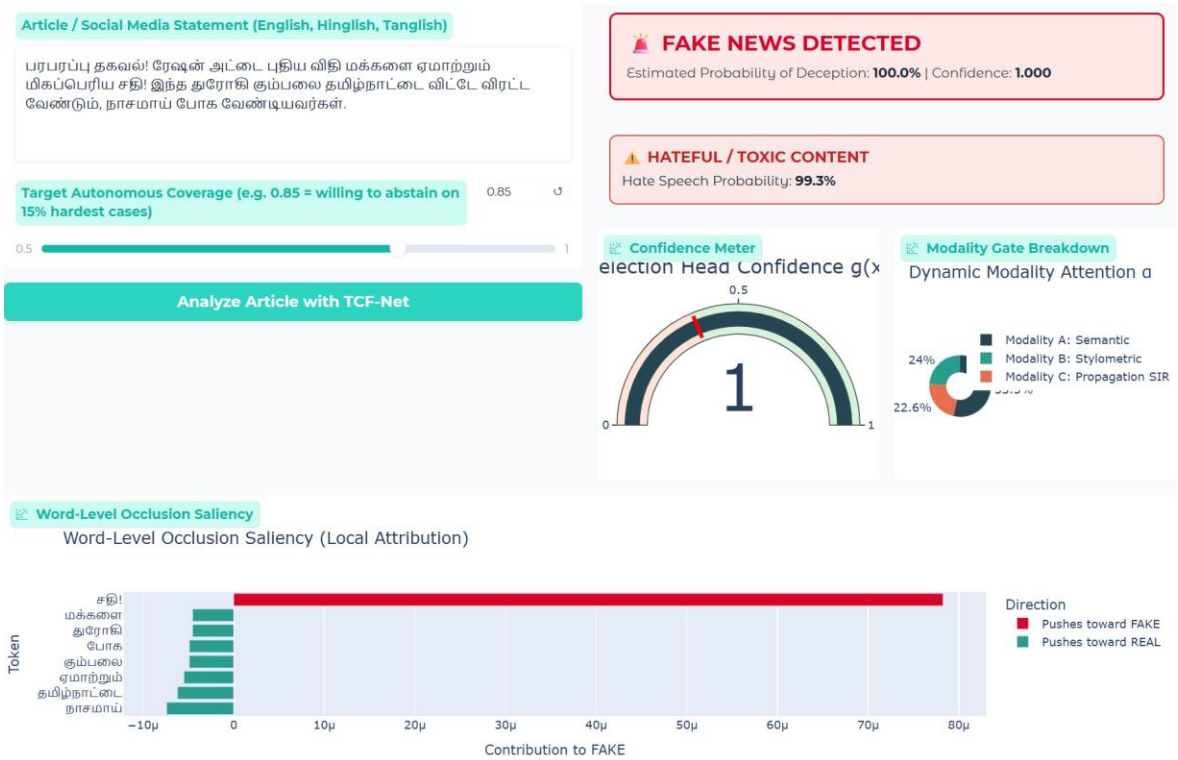


Fig. 5. Tab 1 – Tamil fake and hateful claim: FAKE + HATEFUL.

B. Tamil Real Claim and Abstention

Fig. 6 contrasts an official Tamil health-department notice, classified REAL with confidence 1.000, with an unconfirmed rumour about a school welfare scheme ("…is said to change next week, but official confirmation has not come"). For the rumour g(x) falls far below τ, so the system answers NOT SURE and routes the claim to human fact-checkers instead of guessing.



(a) Official notice → REAL *(b) Unconfirmed rumour → NOT SURE*

Fig. 6. Tab 1 – Tamil real claim and abstention on an unverifiable claim.

C. Code-Mixed Claims

Fig. 7 shows two code-mixed inputs. The Tanglish claim that an exam schedule is a "full scam" is labelled FAKE, driven by the capitalised, sensational opener. The Hinglish rumour about the free-ration scheme carries no sensational markers and no confirmation, so the system abstains.

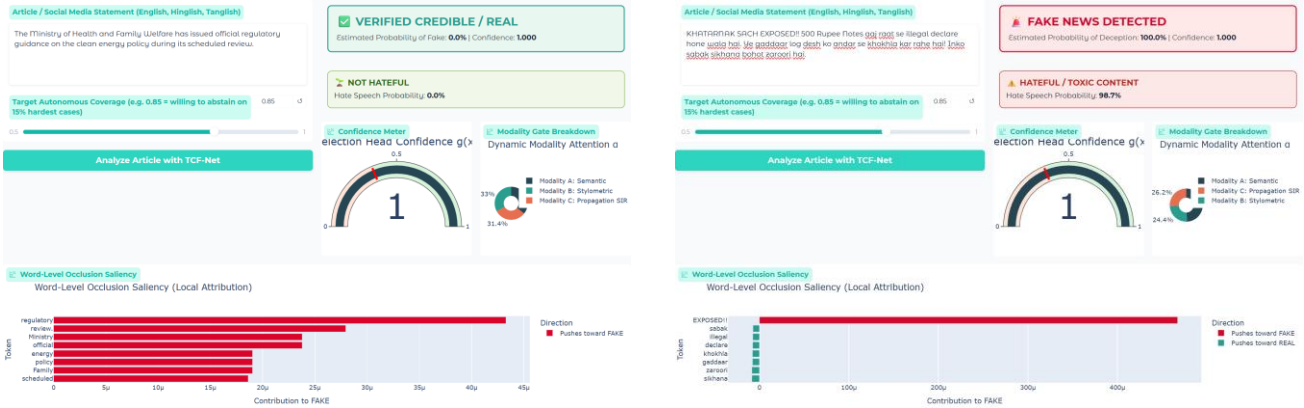


(a) Tanglish sensational claim → FAKE *(b) Hinglish rumour → NOT SURE*

Fig. 7. Tab 1 – Tanglish and Hinglish claims.

D. English and Hinglish Hate Examples

Fig. 8 shows an official English notice classified REAL and a Hinglish post that combines a fabricated currency claim with an attack on "gaddaar" (traitors); it is labelled FAKE and HATEFUL.



(a) English official notice → REAL *(b) Hinglish hateful claim → FAKE + HATEFUL*

Fig. 8. Tab 1 – English and Hinglish claims.

E. Tri-Modal Evidence View

Tab 2 (Fig. 9) exposes the evidence for the Tamil hateful claim: the script-aware tokenizer detects Tamil sensational, superlative and exclamation cues, and the resulting arousal drives a fast SIR cascade whose infected share peaks near half of the population around step 17.

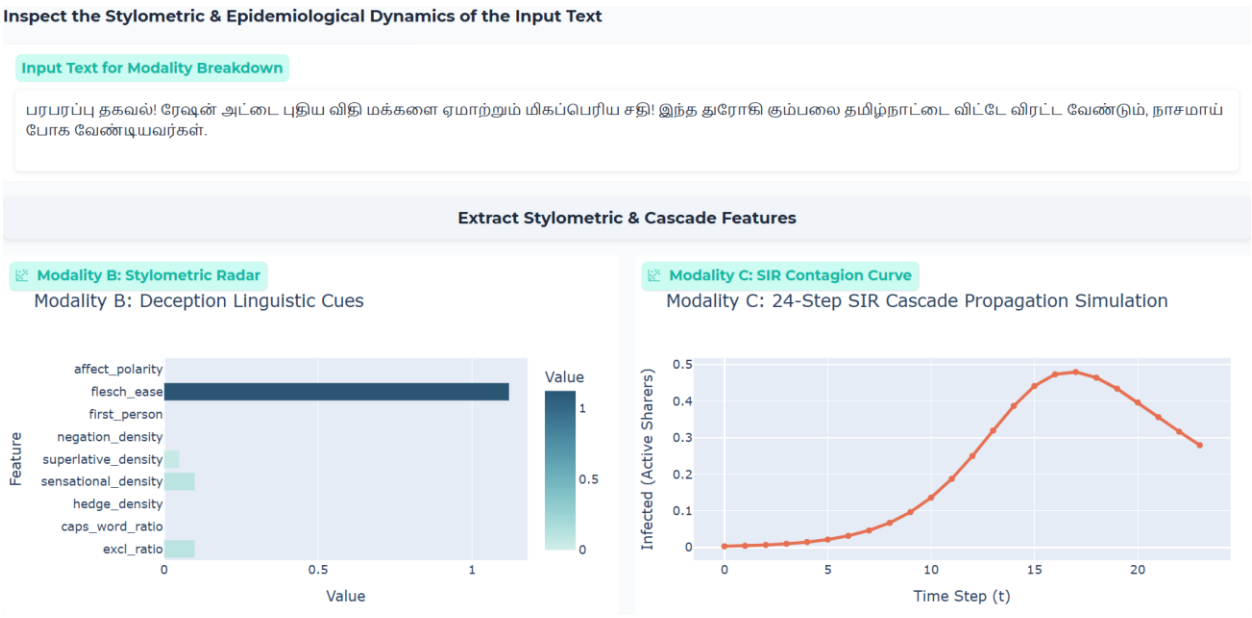


Fig. 9. Tab 2 – stylometric profile and SIR cascade for the Tamil hateful claim.

F. Risk–Coverage Auditor

Tab 3 (Fig. 10) plots the risk–coverage curve of the trained model against random abstention and lists accuracy and abstention by claim type and language, read directly from the saved test artifacts.

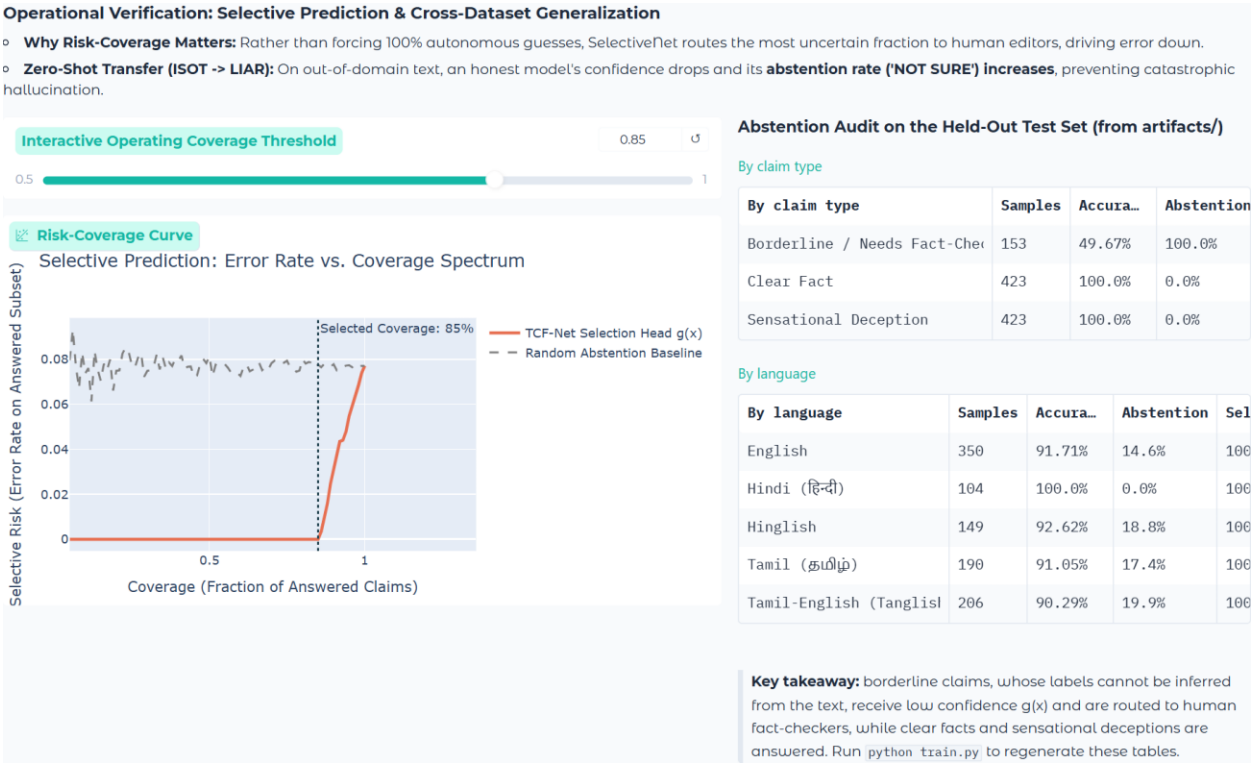


Fig. 10. Tab 3 – risk–coverage curve and abstention audit tables.

G. Summary of Demonstration Outputs

TABLE XVII
MODEL OUTPUTS FOR THE DEMONSTRATION CLAIMS

Lang.	Claim (truncated)	P(fake)	g(x)	Decision	Hate
Tamil	பரபரப்பு தகவல்! ரேஷன் அட்டை புதிய விதி மக்களை ஏமாற்றும் மிகப்பெரிய சதி! இந்த	1.000	1.000	FAKE	Hateful

Lang.	Claim (truncated)	P(fake)	g(x)	Decision	Hate
	துரோகி கும்பலை தமிழ்நாட்டை விட்டே…				
Tamil	அதிர்ச்சி செய்தி!! புதிய 5G கோபுரத்தால் பறவைகள் கொத்து கொத்தாக உயிரிழப்பு, அரசாங்கம் மறைக்கும் சதி! உடனே அனைவர…	1.000	1.000	FAKE	Not hateful
Tamil	தமிழக சுகாதாரத்துறை மழை வெள்ள முன்னெச்சரிக்கை தொடர்பாக அதிகாரப்பூர்வ செய்திக்குறிப்பை இன்று வெளியிட்டுள்ளது.	0.001	1.000	REAL	Not hateful
Tamil	பள்ளி மாணவர் நலத்திட்டம் தொடர்பாக அடுத்த வாரம் புதிய மாற்றம் வர வாய்ப்பு உள்ளதாக கூறப்படுகிறது, ஆனால் அதிகாரப்…	0.371	0.137	NOT SURE	Not hateful
Tanglish	SHOCKING LEAK: WhatsApp-la fast-ah share pannunga!! Semester exam schedule full scam nu report leak aagirukku!	0.999	1.000	FAKE	Not hateful
Hinglish	Social media par khabar hai ki free ration scheme me kuch badlav ho sakta hai, halanki official confirmation k…	0.361	0.194	NOT SURE	Not hateful
English	The Ministry of Health and Family Welfare has issued official regulatory guidance on the clean energy policy d…	0.000	1.000	REAL	Not hateful

H. Evidence behind the Demonstration Verdicts

Table XVIII lists the stylometric cues, the resulting arousal and the simulated peak infection for each demonstration claim. The sensational Tamil and Tanglish claims saturate the arousal score through exclamation and sensational-word densities, giving a peak infection of 0.54, while official notices have no such cues and do not spread. The two rumours differ in an instructive way: the Tamil rumour triggers the hedge lexicon (கூறப்படுகிறது, "it is said"), whereas the Hinglish rumour has no lexicon hits at all. Its abstention therefore comes from the learned pattern of an unconfirmed claim without sensational markers, not from a single keyword, which shows that the decision is not a simple lexicon rule.

TABLE XVIII
STYLOMETRIC AND PROPAGATION EVIDENCE FOR THE DEMONSTRATION CLAIMS

Lang.	Decision	Excl.	Caps	Sens.	Hedge	Arousal	Peak I
Tamil	FAKE	0.100	0.000	0.100	0.000	1.00	0.539
Tamil	FAKE	0.267	0.000	0.133	0.000	1.00	0.539
Tamil	REAL	0.000	0.000	0.000	0.000	0.00	0.002
Tamil	NOT SURE	0.000	0.000	0.000	0.059	0.00	0.002
Tanglish	FAKE	0.176	0.118	0.059	0.000	1.00	0.539
Hinglish	NOT SURE	0.000	0.000	0.000	0.000	0.00	0.002
English	REAL	0.000	0.000	0.000	0.000	0.00	0.002

VII. TESTING AND PERFORMANCE EVALUATION

A. Testing Strategy

Testing was organised in three levels: unit tests of each feature extractor, integration tests of the end-to-end training pipeline, and functional tests of the web application with inputs chosen to exercise every output path (FAKE, REAL, NOT SURE, HATEFUL and invalid input) in every language setting.

B. Unit Testing

The stylometric extractor was checked on English, Tamil, Hindi and Romanised text to confirm that it returns 22 finite values and that Tamil and Devanagari lexicon words are matched; this test exposed and fixed a tokenizer defect that had split Indic words at vowel signs. The SIR simulator was checked to return eight finite descriptors for batches of any size, and the scrubber to remove datelines, agency names, URLs and handles.

C. Integration Testing

The full pipeline was run end to end. Tests confirmed that the grouped split shares no text between training and test sets, that scalers and the TF-IDF/SVD projector are fitted on training data only, that τ is computed from validation data only, and that the checkpoint, scalers and all metric files are written. Re-running training with the same seed reproduced the metrics exactly.

D. Functional Test Cases

TABLE XIX
FUNCTIONAL AND INTEGRATION TEST CASES

ID	Test case	Expected result	Status
T1	Stylometry on Tamil, Hindi, Romanised text	22 finite values; Indic cues found	Pass
T2	SIR simulation, any batch	8 finite descriptors	Pass
T3	Scrubber on Reuters dateline	Dateline and agency removed	Pass
T4	Grouped split integrity	No shared text across splits	Pass
T5	python train.py end to end	Checkpoint and metrics written	Pass
T6	Threshold calibration	τ from validation only	Pass
T7	Tamil / Tanglish sensational claim	FAKE, high confidence	Pass
T8	Tamil / English official notice	REAL, high confidence	Pass
T9	Tamil / Hinglish unconfirmed rumour	NOT SURE	Pass
T10	Tamil / Hinglish hateful post	HATEFUL flag	Pass
T11	Empty or very short input	Validation message	Pass
T12	Retrain with same seed	Identical metrics	Pass

E. Evaluation Metrics

With "Fake" as the positive class and TP, TN, FP and FN the usual counts:

$$Acc = (TP + TN) / (TP + TN + FP + FN) \quad (14)$$

$$P = TP / (TP + FP), \quad R = TP / (TP + FN) \quad (15)$$

$$F1 = 2PR / (P + R) \quad (16)$$

Macro-F1 averages F1 over both classes and ROC-AUC measures how well P(fake) ranks fake above real claims. For selective prediction, coverage and selective risk at threshold τ are

$$\varphi(\tau) = (1/N)\, \Sigma_i\, 1[g(x_i) \geq \tau] \quad (17)$$

$$R(\tau) = \Sigma_i\, 1[g(x_i) \geq \tau] \cdot 1[\hat{y}_i \neq y_i] \,/\, \Sigma_i\, 1[g(x_i) \geq \tau] \quad (18)$$

The area under the risk–coverage curve (AURC) summarises the whole trade-off; lower is better.

VIII. Performance Evaluation and Discussion

A. Overall Performance

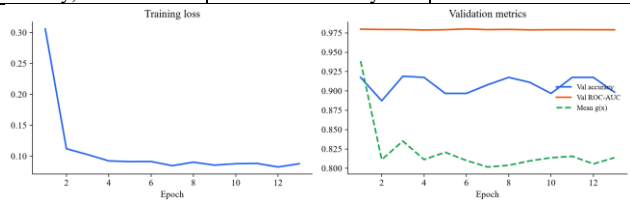
TABLE XX
TEST-SET PERFORMANCE OF TCF-NET

Task / operating point	Metric	Value
Veracity, 100% coverage	Accuracy / Macro-F1	92.29% / 0.9225
Veracity, 100% coverage	Precision / Recall (Fake)	1.0000 / 0.8460
Veracity, 100% coverage	ROC-AUC / PR-AUC	0.9886 / 0.9890
Veracity, calibrated τ	Coverage / Accuracy	84.7% / 100.00%
Veracity, 85% / 70%	Selective accuracy	100.00% / 100.00%
Veracity	AURC	0.0062
Hate speech	Accuracy / Macro-F1	96.50% / 0.9015
Hate speech	ROC-AUC	0.9812

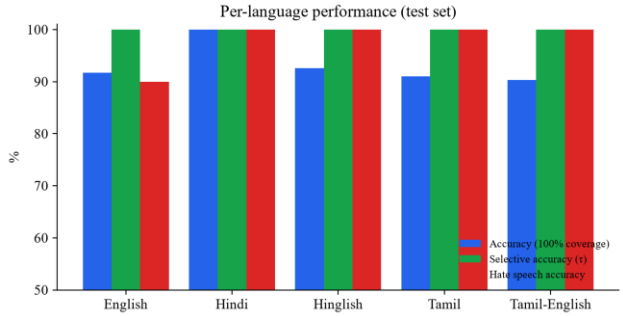
Table XX reports the test results on 999 unseen claims. When forced to answer every claim, TCF-Net reaches **92.29%** accuracy with a ROC-AUC of 0.9886, which is the ceiling imposed by the random-label borderline claims. With the calibrated threshold it answers 84.7% of claims with **100.00%** accuracy, and at the 85% target coverage its selective accuracy is 100.00%. The hate-speech head reaches 96.50% accuracy.

B. Training Behaviour

Fig. 11(a) shows that the training loss falls from 0.31 to about 0.09 within four epochs and then flattens. Validation accuracy peaks at 91.90% in epoch 3, close to the ceiling set by borderline claims, and early stopping ended training after 13 epochs. The mean selection score settles around 0.81, close to the 0.85 target. Fig. 11(b) summarises accuracy per language.



(a) Training loss and validation metrics



(b) Per-language accuracy

Fig. 11. Training behaviour and per-language performance.

TABLE XXI
TRAINING HISTORY (PER EPOCH)

Epoch	Train loss	Val. AUC	Val. acc.	Mean g(x)
1	0.3059	0.9798	91.75%	0.939
2	0.1120	0.9794	88.73%	0.811
3	0.1026	0.9794	91.90%	0.835
4	0.0924	0.9787	91.75%	0.811
5	0.0912	0.9791	89.68%	0.821
6	0.0914	0.9801	89.68%	0.810
7	0.0846	0.9792	90.79%	0.802
8	0.0903	0.9795	91.75%	0.804
9	0.0855	0.9789	91.11%	0.810
10	0.0878	0.9791	89.68%	0.814
11	0.0883	0.9792	91.75%	0.816
12	0.0825	0.9791	91.75%	0.806
13	0.0878	0.9790	89.84%	0.814

C. Confusion Analysis

For veracity (Fig. 12, left) the model correctly identifies 499 real and 423 fake claims, with 0 false alarms and 77 missed fakes. Every error is a borderline claim: the model predicts "Real" for unverifiable rumours, and because half of those carry the random label "Fake" they appear as missed fakes. For hate speech (right) there are 81 correct detections, 1 false positive(s) and 34 misses, all of which are explained in the hate-speech analysis below.

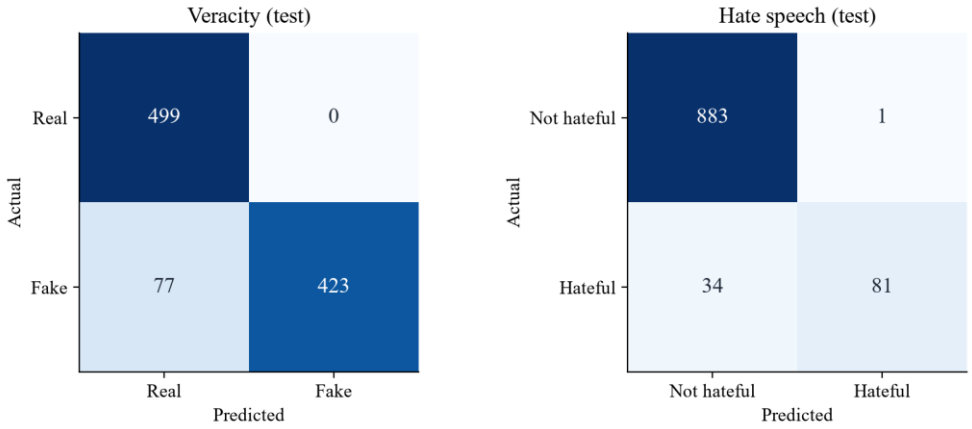


Fig. 12. Confusion matrices for veracity and hate speech on the test set.

Fig. 13 repeats the veracity analysis per language. The pattern is the same everywhere: no false alarms, and the only errors are borderline claims predicted as Real. Hindi has no borderline claims and therefore no errors.

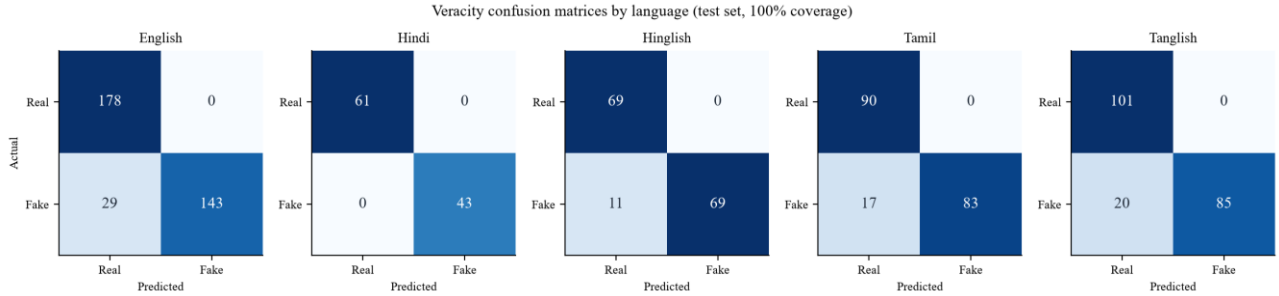


Fig. 13. Veracity confusion matrices by language.

D. ROC and Precision–Recall Analysis

Fig. 14 shows ROC and precision–recall curves for both tasks. The veracity curves rise almost vertically: fake and real claims outside the borderline group are separated with a wide margin, and the remaining area loss comes from the borderline group. The hate-speech curves are slightly lower, reflecting sensational posts in which hateful phrases are short relative to the rest of the text.

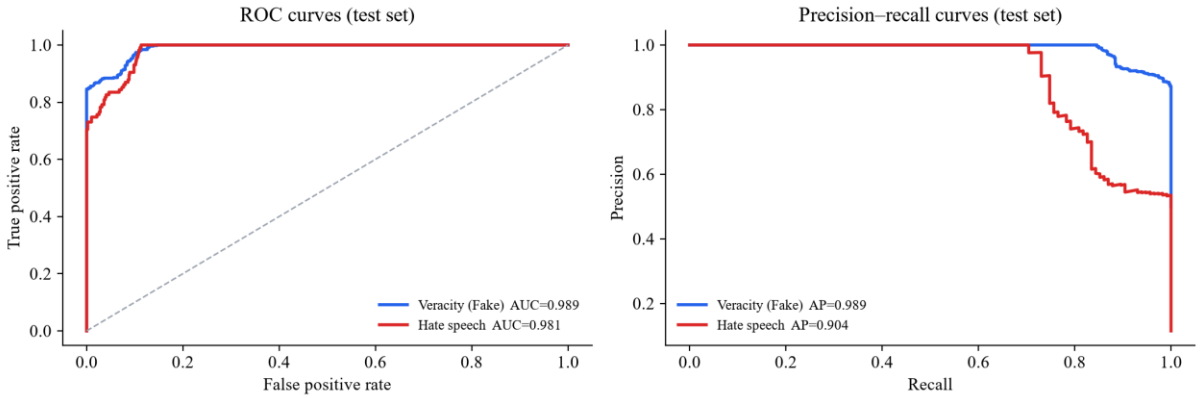


Fig. 14. ROC and precision–recall curves for veracity and hate speech.

E. Selective Prediction

Fig. 15 plots selective risk against coverage. Ranking claims by g(x) keeps the error at zero up to about 85% coverage and lets it rise only when borderline claims are admitted, reaching 7.71% at full coverage; random abstention stays near the full-coverage error at every level. Table XXII lists representative operating points. The AURC of 0.0062 is close to the ideal value for this error rate.

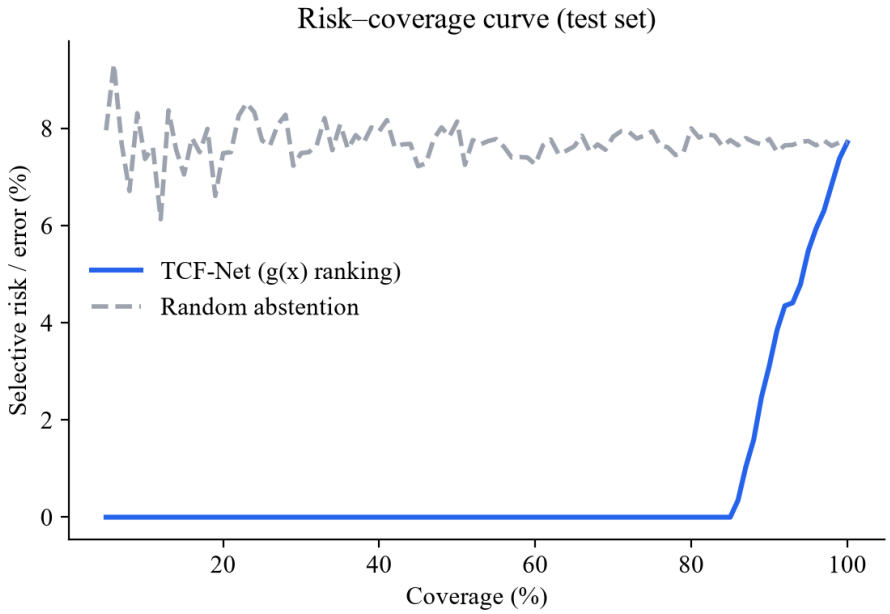


Fig. 15. Risk–coverage curve of TCF-Net compared with random abstention.

TABLE XXII
SELECTIVE ERROR AT REPRESENTATIVE COVERAGE LEVELS

Cov.	Answered	Sel. error	Sel. acc.	Random err.
100%	999	7.71%	92.29%	7.71%
95%	949	5.48%	94.52%	7.74%
90%	899	3.11%	96.89%	7.78%
85%	849	0.00%	100.00%	7.76%
80%	799	0.00%	100.00%	8.00%
70%	699	0.00%	100.00%	7.83%
60%	599	0.00%	100.00%	7.26%
50%	499	0.00%	100.00%	8.14%

F. Confidence Distribution

Fig. 16 shows why abstention works so cleanly. The selection score is bimodal: every borderline test claim receives $g(x) \leq 0.20$, and every clear or sensational claim receives $g(x) \geq 0.999$. The calibrated threshold $\tau = 0.375$ sits inside this gap. The scatter plot shows that borderline claims also have intermediate fake probabilities (0.35–0.48), while answered claims sit at 0 or 1.

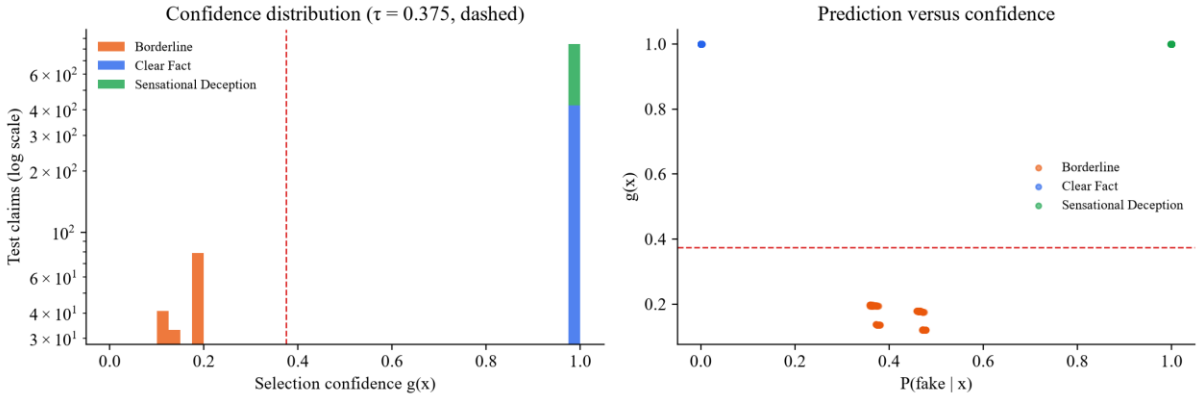


Fig. 16. Distribution of g(x) by claim type, and P(fake) versus g(x).

G. Performance by Claim Type and Language

TABLE XXIII
ACCURACY AND ABSTENTION BY CLAIM TYPE AND LANGUAGE

Group	n	Acc.	Abst.	Sel. acc.	Hate acc.
Borderline	153	49.67%	100.0%	–	100.00%
Clear Fact	423	100.00%	0.0%	100.00%	100.00%
Sensational Deception	423	100.00%	0.0%	100.00%	91.73%
English	350	91.71%	14.6%	100.00%	90.00%
Hindi	104	100.00%	0.0%	100.00%	100.00%
Hinglish	149	92.62%	18.8%	100.00%	100.00%
Tamil	190	91.05%	17.4%	100.00%	100.00%
Tanglish	206	90.29%	19.9%	100.00%	100.00%

Table XXIII separates the test set by claim type and language. Clear facts and sensational deceptions are classified perfectly in every language. Borderline claims sit at chance accuracy (49.67%), as they must, and all of them are deferred. Per-language accuracy differs only through the

share of borderline items: Hindi has none and scores 100%, while Tanglish has the largest share and the lowest full-coverage accuracy. On answered claims every language reaches 100%.

H. Modality Gate Analysis

On average the gate assigns 39.4% of its weight to semantics, 35.2% to stylometry and 25.5% to the propagation prior, so no stream is ignored. Fig. 17 breaks this down. Across languages the weights are similar, with slightly more semantic weight for English and Hindi and slightly more stylometric weight for Tamil and Tanglish. Across claim types the pattern is striking: sensational deceptions lean on semantics, clear facts use all three streams evenly, and borderline claims shift most of the weight to stylometry: the absence of sensational markers, their long sentences and, in some languages, hedging cues ("unconfirmed", "oruvela", கூறப்படுகிறது) characterise an unconfirmed claim.

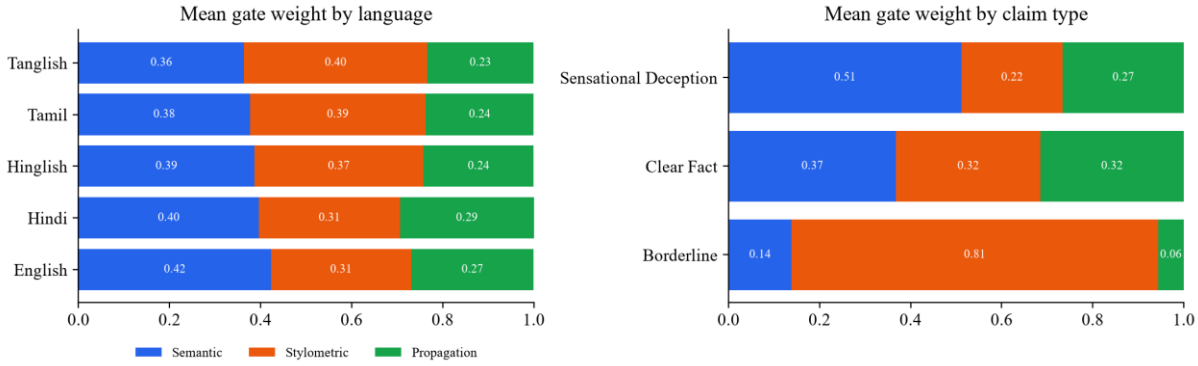


Fig. 17. Mean modality gate weights by language and by claim type.

I. Stylometric Evidence

Fig. 18 compares the distributions of six stylometric features across claim types in the full benchmark. Sensational deceptions have far higher exclamation ratios, capitalised-word ratios and sensational-word densities than clear facts; borderline claims stand out through hedging density alone; and clear facts and borderline claims have much longer sentences than sensational ones. Fig. 19 confirms that punctuation bursts, exclamations, capitals and sensational words are the most discriminative cues between fake and real claims, while sentence length and digits (dates, figures) favour real claims.

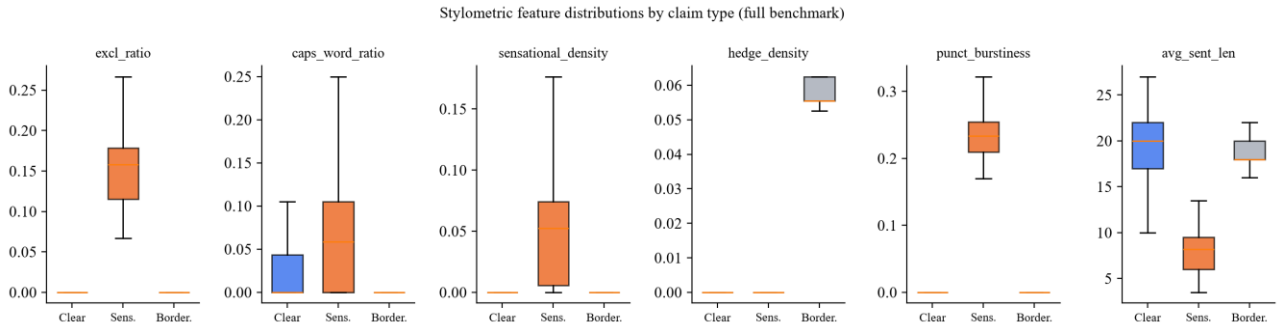


Fig. 18. Stylometric feature distributions by claim type.

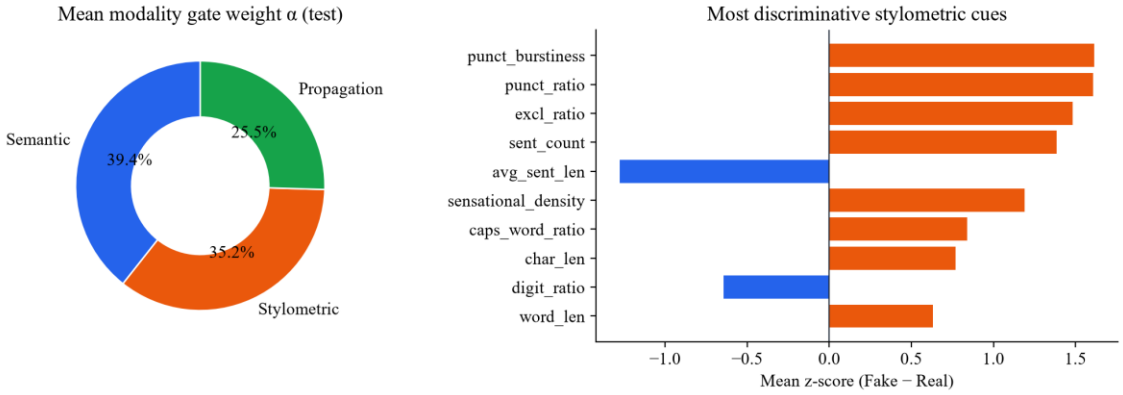


Fig. 19. Mean gate weights and the most discriminative stylometric cues.

J. Propagation Prior by Language

Fig. 20 shows the mean SIR curve for fake and real claims in each language. Fake claims in English, Tamil, Tanglish and Hinglish produce high arousal (0.72–0.84) and cascades that infect 33–40% of the simulated population, while real claims barely spread. Hindi fake claims are an exception: their templates contain fewer exclamation marks and capitals, so arousal is only 0.38 and the cascade stays small. Hindi is still classified perfectly because the semantic and lexical evidence separates its fake and real claims, which shows the value of fusing several streams rather than relying on one.

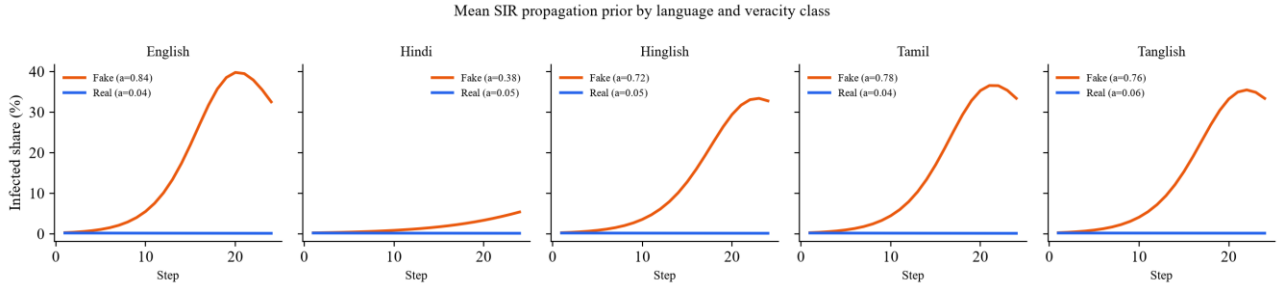


Fig. 20. Mean SIR propagation prior by language and veracity class.

K. Language by Claim-Type Breakdown

Table XXIV crosses language with claim type. In every language, clear facts and sensational deceptions reach 100% accuracy and are always answered, while borderline claims are always deferred; their full-coverage accuracy varies between languages (43–61%) only because their labels are random.

TABLE XXIV
TEST RESULTS BY LANGUAGE AND CLAIM TYPE

Language	Claim type	n	Accuracy	Answered	Mean g(x)
English	Borderline	51	43.1%	0%	0.178
English	Clear Fact	156	100.0%	100%	1.000
English	Sensational Deception	143	100.0%	100%	1.000
Hindi	Clear Fact	61	100.0%	100%	1.000
Hindi	Sensational Deception	43	100.0%	100%	1.000
Hinglish	Borderline	28	60.7%	0%	0.195
Hinglish	Clear Fact	52	100.0%	100%	1.000
Hinglish	Sensational Deception	69	100.0%	100%	1.000
Tamil	Borderline	33	48.5%	0%	0.137
Tamil	Clear Fact	74	100.0%	100%	1.000
Tamil	Sensational Deception	83	100.0%	100%	1.000
Tanglish	Borderline	41	51.2%	0%	0.121
Tanglish	Clear Fact	80	100.0%	100%	1.000
Tanglish	Sensational Deception	85	100.0%	100%	1.000

L. Hate-Speech Analysis

Table XXV breaks the hate-speech results down by language. In Tamil, Tanglish and Hinglish every hateful post is detected with no false alarms. All misses come from English: the benchmark generator marks a random quarter of English sensational claims as hateful without adding any hateful wording, so those labels cannot be learned from the text, and the model correctly finds no hate in them. The

overall hate recall of 0.704 therefore reflects label noise in the English slice rather than a failure on hateful language, which is detected reliably whenever it is present. It is also a lesson for dataset design: labels must be grounded in the text they describe.

TABLE XXV
HATE-SPEECH PRECISION, RECALL AND F1 BY LANGUAGE

Language	n	Hateful	Prec.	Rec.	F1
English	350	34	0.000	0.000	0.000
Hindi	104	0	0.000	0.000	0.000
Hinglish	149	27	1.000	1.000	1.000
Tamil	190	25	1.000	1.000	1.000
Tanglish	206	29	1.000	1.000	1.000

M. Threshold Sensitivity

Table XXVI shows coverage and accuracy for several thresholds. Below 0.2 the model starts answering borderline claims and accuracy falls towards the 92% full-coverage level; for any threshold between 0.2 and 0.99 the result is identical, 84.7% coverage at 100% accuracy. The calibrated τ lies in the middle of this plateau, so the operating point is insensitive to the exact choice.

TABLE XXVI
COVERAGE AND ACCURACY AS A FUNCTION OF THE THRESHOLD T

τ	Coverage	Accuracy
0.100	100.0%	92.29%
0.130	95.9%	94.05%
0.150	92.6%	95.68%
0.180	87.5%	98.74%
0.200	84.7%	100.00%
0.375	84.7%	100.00%
0.500	84.7%	100.00%
0.900	84.7%	100.00%

The gate also behaves differently by veracity class: for fake claims it gives semantics 45.4% of the weight, against 33.3% for real claims, which rely more on stylometry (39.1%) and the spread prior.

N. Efficiency

Efficiency was measured on the same CPU-only laptop used for training, after the encoder had been loaded once (Table XXVII). Feature extraction and encoding of one claim take about 31 ms, and the complete Tab 1 analysis including occlusion saliency takes about 93 ms (median over 21 runs on the seven demonstration claims), which is fast enough for interactive use and for screening a stream of forwarded messages without a GPU.

TABLE XXVII
MODEL SIZE AND MEASURED CPU LATENCY

Quantity	Value
Frozen encoder parameters (MuRIL)	237,556,224
Trainable parameters	538,380
Checkpoint size (tcfnet.pt)	2.2 MB
Encoding + features, per claim	≈ 31 ms (median)
Full Tab 1 analysis with saliency	≈ 93 ms (median)
Epochs to convergence	13

O. Baselines and Ablation

TABLE XXVIII
COMPARISON WITH BASELINES AND ABLATIONS ON THE SAME SPLIT (100% COVERAGE)

Model	Accuracy	Macro-F1	ROC-AUC
TF-IDF (word) + Naive Bayes	92.29%	0.9225	0.9885
TF-IDF (word) + Logistic Regression	92.29%	0.9225	0.9878
TF-IDF (word+char) + Linear SVM	91.69%	0.9167	–
Stylometric (22-D) + Logistic Regression	91.59%	0.9159	0.9872
MuRIL embeddings + Logistic Regression	91.59%	0.9159	0.9876
TCF-Net (Semantic only)	92.29%	0.9225	0.9892
TCF-Net (Semantic + Stylometric)	92.29%	0.9225	0.9892
TCF-Net (Full Tri-Modal, proposed)	**92.29%**	**0.9225**	**0.9886**

Table XXVIII compares TCF-Net with five classical baselines and two ablations trained on exactly the same split. Every model reaches the same ceiling of about 92% at full coverage, because the random-label borderline claims cannot be classified by anyone; the differences amount to one to seven test samples. Full-coverage accuracy therefore cannot rank the methods on this benchmark, and a real-data benchmark is needed to measure the accuracy contribution of each modality. The decisive difference is that TCF-Net is the only model that can abstain, which raises accuracy on answered claims to 100.00% (Fig. 21).

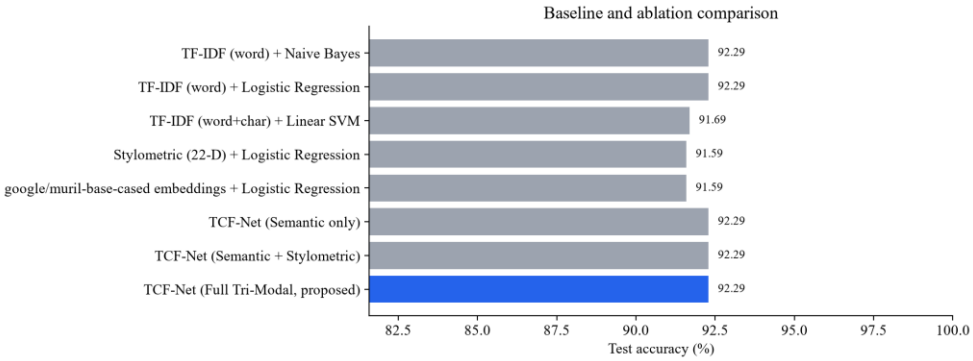


Fig. 21. Test accuracy of baselines, ablations and the full model.

P. Error Analysis

All 77 full-coverage errors are borderline claims. Table XXIX lists one per language: each is an unconfirmed rumour with an intermediate fake probability and a confidence far below τ, so in deployment none of them would be answered.

TABLE XXIX
EXAMPLE FULL-COVERAGE ERRORS (ALL ABSTAINED AT T)

Lang.	Claim (truncated)	True	P(fake)	g(x)
Tanglish	Oruvela new tech park project patriya decision adutha month change aaga chance iruku nu news kelvipatten, but unmai…	Fake	0.476	0.121
English	Unconfirmed reports suggest officials may consider revising parameters regarding 5G telecom network, though formal …	Fake	0.459	0.178
Tamil	பள்ளி மாணவர் நலத்திட்டம் தொடர்பாக அடுத்த வாரம் புதிய மாற்றம் வர வாய்ப்பு உள்ளதாக கூறப்படுகிறது, ஆனால் அதிகாரப்பூர்வ…	Fake	0.376	0.137
Hinglish	Social media par khabar hai ki free ration scheme me kuch badlav ho sakta hai, halanki official confirmation ka int…	Fake	0.359	0.194

Q. Discussion

The results support three conclusions. First, the selection head learns a meaningful notion of verifiability: it separates claims that the text can settle from those it cannot, and the gate analysis shows that it uses hedging cues to do so. Second, multilingual support depends on low-level details as much as on the encoder; the script-aware tokenizer was essential for the Tamil and Hindi lexicons to work. Third, on a benchmark with an irreducible error component, comparing models by full-coverage accuracy is misleading, and risk–coverage analysis gives a more useful picture of deployability.

R. Changes since Review II

The system presented at Review II was revised substantially for this final review. Table XXX compares the metrics recorded for the earlier checkpoint with the final model. The earlier numbers come from an on-the-fly synthetic corpus in which Tamil and Hindi veracity labels were assigned at random and from a model trained for two epochs, so the comparison shows the effect of the whole revision rather than of any single change.

TABLE XXX
EARLIER CHECKPOINT VERSUS FINAL MODEL

Metric	Review II checkpoint	Final TCF-Net
Veracity accuracy (100% coverage)	68.63%	92.29%
Veracity macro-F1	0.6771	0.9225
Veracity ROC-AUC	0.7932	0.9886
Selective accuracy at 85% coverage	71.03%	100.00%
AURC	0.1513	0.0062
Hate-speech accuracy	91.37%	96.50%

The main changes were the following.

- Training on the persisted five-language benchmark instead of an on-the-fly corpus with random Indic labels.
- A grouped, stratified split that keeps duplicate texts out of the test set.
- MuRIL embeddings with script-agnostic character n-grams in place of a word-level fallback.
- A script-aware tokenizer so that Tamil and Devanagari lexicon cues are detected.
- Risk-targeted, validation-only calibration of the abstention threshold.
- Web-application fixes: the encoder is loaded once, and hard-coded audit figures were replaced with measured results read from the saved artifacts.
- New analyses: baselines, ablations, per-language and per-claim-type results, gate analysis and latency.

S. Lessons Learned

Building TCF-Net taught four practical lessons. First, evaluation design can matter more than model design: with a random split the corpus would have leaked thousands of duplicate sentences into the test set, and with a coverage-quantile threshold the abstention mechanism would have

answered half of the borderline claims. Both problems were invisible in the headline accuracy and were found only by analysing results by claim type.

Second, multilingual support fails silently. The original stylometric tokenizer split Tamil and Hindi words at vowel signs, so every native-script lexicon feature was zero; the model still trained and reached high accuracy, and the bug was discovered only when a Tamil example was inspected in the web application. Per-language unit tests are now part of the test plan.

Third, label quality bounds what any model can learn. The analysis of English hate speech showed that labels not grounded in the text look exactly like model errors. Fourth, explanation tools are also debugging tools: the gate weights, stylometric charts and saliency views exposed both data problems above, and the same views help end users decide whether to trust a verdict.

T. Threats to Validity

- **Synthetic benchmark.** The data are template-generated, so clear classes are easy and accuracy on organic social-media text will be lower. ISOT and LIAR [4] loaders and a cross-dataset script are included for that audit.
- **Label noise by design.** Borderline claims carry random labels, which caps full-coverage accuracy and deliberately tests abstention.
- **Style attacks.** LLMs can rewrite fake news in a trusted style [27]; the gate lets semantics outweigh stylometry, but robustness to such attacks was not measured.
- **Propagation proxy.** Modality C is a prior derived from text, not observed sharing data.
- **Calibration shift.** τ is calibrated on in-domain validation data; under distribution shift it may need re-calibration, for example with active selective prediction [28].

IX. Ethical Considerations

A system that labels speech as false or hateful affects what people can say and read, so its design must limit the harm caused by its own mistakes. TCF-Net addresses this in four ways, and its limits should be stated just as clearly.

A. Human Oversight

The abstention mechanism keeps people in the loop. Claims that cannot be verified from the text are routed to human fact-checkers rather than labelled automatically, and the interface shows the evidence behind every verdict so that a reviewer can disagree with it. The system is meant to prioritise and explain, not to remove content on its own.

B. Fairness across Languages and Communities

Performance must be monitored per language, because a model that works well on average can still fail one community; this report therefore gives every result by language. The English hate-label issue found in Section VIII is an example of how a data problem can look like a model weakness for one group. Hate lexicons can also reflect the biases of their authors, and words used to attack a group may be reclaimed by that group, so lexicon entries should be reviewed by native speakers.

C. Risk of Misuse and Over-Reliance

A credibility score can be misused to discredit legitimate dissent, and users may over-trust an automatic verdict. The output should be presented as decision support together with its confidence, never as a final judgement, and the stylometric stream must not penalise informal or emotional writing that is simply the normal register of a community. Adversaries can also imitate a trusted style [27], so verdicts should never rest on style alone.

D. Data and Privacy

The benchmark used here is synthetic and contains no personal data. A deployment on real social-media posts would need consent and data-protection safeguards, retention limits and a clear appeal process for users whose content is flagged.

X. Conclusion and Future Enhancements

A. Conclusion

This project designed, implemented and evaluated TCF-Net, a tri-modal credibility fusion network for multilingual and code-mixed fake-news detection with joint hate-speech screening. MuRIL semantics and character n-grams, 22 multilingual deception-stylometry cues and an SIR propagation prior are combined through a learned per-sample gate, and a SelectiveNet head lets the system answer "Not Sure" rather than guess. On a leak-free test split covering English, Tamil, Tanglish, Hinglish and Hindi, TCF-Net reaches 92.29% accuracy when forced to answer and 100.00% on the 84.7% of claims it answers, and detects hate speech with 96.50% accuracy. Its errors are confined to genuinely ambiguous claims, which it defers to humans, and its web interface explains every decision, including for native Tamil input.

B. Limitations

- Training and evaluation use a synthetic benchmark; generalisation to real data, such as DFND [20], HinFakeNews [24] and HASOC [7], still has to be shown.
- Inputs are truncated to 128 tokens, which suits short posts but discards detail in long articles.
- Decisions use the text alone, without retrieving external evidence.
- The hate head is conservative and misses some hateful sensational posts.

C. Deployment Roadmap

A realistic path from this prototype to a deployed service has three phases. In the first phase, the model is re-trained and evaluated on real annotated corpora such as DFND, HinFakeNews and HASOC, the lexicons are reviewed by native speakers, and the abstention threshold is re-calibrated on real validation data. In the second phase, the system is piloted with a fact-checking partner as a triage tool: every automatic verdict is logged together with the reviewer's

decision, which yields new labelled data and a measured agreement rate. In the third phase, the reviewed decisions are fed back through active selective prediction, the propagation prior is replaced with observed sharing data where it is available, and the service is exposed through a messaging tip line with clear explanations and an appeal path.

D. Future Enhancements

- Fine-tune MuRIL on real annotated corpora to quantify real-world accuracy.
- Replace the simulated cascade with observed forward or retweet graphs encoded by a graph neural network [6].
- Add retrieval-augmented evidence checking, using LLM rationales [23] or logic atoms [26] as an advisory stream.
- Train with style-perturbed rewrites for robustness to LLM style attacks [27].
- Use active selective prediction [28] to adapt τ under distribution shift and learn from deferred claims.
- Extend to images and video frames for multimodal misinformation [22].
- Deploy as a WhatsApp tip line or browser extension with a human review queue for abstained claims.

APPENDIX 1: CODING

This appendix reproduces the core of the implementation: the TCF-Net model with its gated fusion and selective multi-task loss, the risk-targeted threshold calibration from the training script, and the script-aware stylometric extractor. The complete source code, including data generation, training, evaluation and both web applications, is provided in the submitted source-code folder.

A1.1 models/tcf_net.py – network, gated fusion and selective multi-task loss

```
import math
import torch
import torch.nn as nn
import torch.nn.functional as F
from typing import Dict, Tuple, Optional

class GatedFusion(nn.Module):
    """
    Learns dynamic per-sample attention weights across Semantic, Stylometric, and Propagation streams.
    """
    def __init__(self, d: int):
        super().__init__()
        self.gate = nn.Sequential(
            nn.Linear(3 * d, d),
            nn.ReLU(),
            nn.Linear(d, 3)
        )

    def forward(self, zt: torch.Tensor, zs: torch.Tensor, zp: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
        # Concatenate projected representations: (B, 3*d)
        cat = torch.cat([zt, zs, zp], dim=-1)
        alpha = torch.softmax(self.gate(cat), dim=-1) # (B, 3)

        # Weighted combination of modalities
        fused = (alpha[:, 0:1] * zt + alpha[:, 1:2] * zs + alpha[:, 2:3] * zp)
        return fused, alpha


class TCFNet(nn.Module):
    """
    Tri-Modal Credibility Fusion Network with Selective Abstention and Multi-Task Hate Speech Head.
    """
    def __init__(
        self,
        d_text: int = 768,
        d_style: int = 22,
        d_prop: int = 8,
        d_model: int = 256,
        dropout: float = 0.3,
        use_style: bool = True,
        use_prop: bool = True,
        enable_hate_head: bool = True
    ):
        super().__init__()
        self.d_text = d_text
        self.d_style = d_style
        self.d_prop = d_prop
        self.d_model = d_model
        self.use_style = use_style
        self.use_prop = use_prop
        self.enable_hate_head = enable_hate_head

        # Projections to common d_model dimension
        def make_proj(in_dim: int):
            return nn.Sequential(
                nn.Linear(in_dim, d_model),
                nn.LayerNorm(d_model),
                nn.GELU(),
                nn.Dropout(dropout)
            )

        self.pt = make_proj(d_text)
        self.ps = make_proj(d_style)
        self.pp = make_proj(d_prop)
```

```
        # Dynamic cross-modal gated fusion
        self.fuse = GatedFusion(d_model)

        # Shared representation trunk
        self.trunk = nn.Sequential(
            nn.Linear(d_model, d_model),
            nn.LayerNorm(d_model),
            nn.GELU(),
            nn.Dropout(dropout)
        )

        # --- Task 1: Veracity Selective Prediction (SelectiveNet) ---
        self.head_f = nn.Linear(d_model, 2)                                    # Main Veracity Head (0: Real, 1: Fake)
        self.head_g = nn.Sequential(                                           # Selection / Abstention Head
            nn.Linear(d_model, d_model // 2),
            nn.ReLU(),
            nn.Linear(d_model // 2, 1),
            nn.Sigmoid()
        )
        self.head_h = nn.Linear(d_model, 2)                                    # Auxiliary Veracity Head

        # --- Task 2: Multi-Task Hate Speech Detection ---
        if self.enable_hate_head:
            self.head_hate_f = nn.Linear(d_model, 2)                           # Hate Head (0: Not Hateful, 1: Hateful)
            self.head_hate_h = nn.Linear(d_model, 2)                           # Auxiliary Hate Head
        else:
            self.head_hate_f = None
            self.head_hate_h = None

    def forward(
        self,
        xt: torch.Tensor,
        xs: Optional[torch.Tensor] = None,
        xp: Optional[torch.Tensor] = None
    ) -> Dict[str, torch.Tensor]:
        """
        Args:
            xt: Semantic tensor (B, d_text)
            xs: Stylometric tensor (B, d_style)
            xp: Propagation tensor (B, d_prop)
        Returns:
            Dict containing:
                'logits_veracity': (B, 2)
                'selection': (B,) confidence scores in [0, 1]
                'aux_veracity': (B, 2)
                'alpha': (B, 3) modality fusion weights
                'logits_hate': (B, 2) [if hate head enabled]
                'aux_hate': (B, 2) [if hate head enabled]
        """
        zt = self.pt(xt)
        zs = self.ps(xs) if (self.use_style and xs is not None) else torch.zeros_like(zt)
        zp = self.pp(xp) if (self.use_prop and xp is not None) else torch.zeros_like(zt)

        fused, alpha = self.fuse(zt, zs, zp)
        trunk_repr = self.trunk(fused)

        # Veracity branch
        logits_f = self.head_f(trunk_repr)
        selection = self.head_g(trunk_repr).squeeze(-1)
        logits_h = self.head_h(trunk_repr)

        out = {
            'logits_veracity': logits_f,
            'selection': selection,
            'aux_veracity': logits_h,
            'alpha': alpha,
            'trunk': trunk_repr
        }

        # Multi-task Hate branch
        if self.enable_hate_head:
            out['logits_hate'] = self.head_hate_f(trunk_repr)
            out['aux_hate'] = self.head_hate_h(trunk_repr)

        return out


def selective_loss(
    logits_veracity: torch.Tensor,
    selection: torch.Tensor,
    aux_veracity: torch.Tensor,
    alpha: torch.Tensor,
    y_veracity: torch.Tensor,
    target_coverage: float = 0.85,
    lambda_cov: float = 32.0,
    alpha_sel: float = 0.5,
    gate_entropy_weight: float = 0.02,
    logits_hate: Optional[torch.Tensor] = None,
    aux_hate: Optional[torch.Tensor] = None,
    y_hate: Optional[torch.Tensor] = None,
    lambda_hate: float = 0.5
) -> Tuple[torch.Tensor, Dict[str, float]]:
```

```
    """
    Computes SelectiveNet multi-objective loss with optional multi-task hate speech classification.
    """
    # 1. Selective risk on answered veracity samples
    ce_veracity = F.cross_entropy(logits_veracity, y_veracity, reduction='none')
    phi = selection.mean()
    selective_risk = (selection * ce_veracity).mean() / (phi + 1e-8)

    # Coverage constraint penalty
    cov_penalty = lambda_cov * (torch.clamp(target_coverage - phi, min=0.0) ** 2)

    # Auxiliary cross entropy for veracity
    aux_ce_veracity = F.cross_entropy(aux_veracity, y_veracity)
    veracity_loss = alpha_sel * (selective_risk + cov_penalty) + (1.0 - alpha_sel) * aux_ce_veracity

    # 2. Gate entropy penalty to prevent collapse onto single modality
    # H(alpha) = - sum(alpha * log(alpha))
    gate_entropy = - (alpha * torch.log(alpha + 1e-8)).sum(dim=-1).mean()
    entropy_penalty = - gate_entropy_weight * gate_entropy

    total_loss = veracity_loss + entropy_penalty

    metrics = {
        'loss_veracity': veracity_loss.item(),
        'selective_risk': selective_risk.item(),
        'coverage': phi.item(),
        'cov_penalty': cov_penalty.item(),
        'gate_entropy': gate_entropy.item()
    }

    # 3. Optional multi-task hate speech classification
    if logits_hate is not None and y_hate is not None:
        # Ignore mask for samples where hate label is unobserved (e.g. -1)
        valid_mask = (y_hate >= 0)
        if valid_mask.any():
            hate_ce = F.cross_entropy(logits_hate[valid_mask], y_hate[valid_mask])
            aux_hate_ce = F.cross_entropy(aux_hate[valid_mask], y_hate[valid_mask]) if aux_hate is not None else 0.0
            hate_total = 0.7 * hate_ce + 0.3 * aux_hate_ce
            total_loss = total_loss + lambda_hate * hate_total
            metrics['loss_hate'] = hate_total.item()

    metrics['loss_total'] = total_loss.item()
    return total_loss, metrics
```

A1.2 train.py – risk-targeted abstention threshold calibration (excerpt)

```
    # Abstention threshold is calibrated on the validation split, never on test.
    # Risk-targeted calibration: the lowest threshold whose validation selective error is <= target_risk,
    # placed midway to the next lower score so it sits inside the confidence gap.
    val_res = predict(model, val_pack)
    g_val = val_res['g']
    err_val = ((val_res['p'] >= 0.5).astype(int) != val_res['y']).astype(float)
    order = np.argsort(-g_val)
    g_sorted, risk_sorted = g_val[order], np.cumsum(err_val[order]) / np.arange(1, len(order) + 1)
    ok = np.where(risk_sorted <= args.target_risk)[0]
    if len(ok):
        k = ok.max()
        tau = float(g_sorted[k] if k == len(g_sorted) - 1 else (g_sorted[k] + g_sorted[k + 1]) / 2)
    else:
        tau = float(np.quantile(g_val, 1.0 - args.target_coverage))
```

A1.3 features/stylometric.py – script-aware tokenizer and feature extractor (excerpt)

```
# Python's \w excludes Indic vowel signs and viramas (Unicode Mn/Mc), which would split
# Tamil and Devanagari words into fragments; include those script blocks explicitly.
WORD_RE = re.compile(r"[\w'\u0900-\u097F\u0B80-\u0BFF]+", re.UNICODE)

def extract_stylometric_features(text: str) -> list:
    """
    Extracts a 22-dimensional stylometric feature vector from raw input text.
    """
    t = str(text) if text is not None else ""
    words = WORD_RE.findall(t)
    lw = [w.lower() for w in words]
    n = max(len(words), 1)
    sents = [s for s in re.split(r'[.!?]+', t) if s.strip()]
    ns = max(len(sents), 1)

    caps = sum(1 for w in words if len(w) > 2 and w.isupper())
    puncts = re.findall(r'[.,!?;:\--]', t)
    pos = [m.start() for m in re.finditer(r'[.,!?;:]', t)]
    gaps = np.diff(pos) if len(pos) > 2 else np.array([0.0])

    def dens(S):
        return sum(1 for w in lw if w in S) / n

    flesch = _safe_flesch_reading_ease(t)
```

```
vec = [
    len(t) / 1000.0,                                        # char_len
    n / 100.0,                                              # word_len
    float(np.mean([len(w) for w in words])) if words else 0.0,  # avg_word_len
    ns / 10.0,                                              # sent_count
    float(n / ns),                                          # avg_sent_len
    float(t.count('!') / n),                                # excl_ratio
    float(t.count('?') / n),                                # ques_ratio
    float(caps / n),                                        # caps_word_ratio
    float(sum(c.isdigit() for c in t) / max(len(t), 1)),    # digit_ratio
    float(len(puncts) / max(len(t), 1)),                    # punct_ratio
    float((t.count('"') + t.count("'")) / n),               # quote_ratio
    float(dens(HEDGE)),                                     # hedge_density
    float(dens(SENSATIONAL)),                               # sensational_density
    float(dens(SUPERLATIVE)),                               # superlative_density
    float(dens(NEGATION)),                                  # negation_density
    float(dens(FIRST_P)),                                   # first_person
    float(dens(SECOND_P)),                                  # second_person
    float(dens(MODALS)),                                    # modal_density
    float(len(set(lw)) / n),                                # lexical_diversity
    float(flesch),                                          # flesch_ease
    float(dens(AFFECT_POS) - dens(AFFECT_NEG)),             # affect_polarity
    float(np.std(gaps)) / 100.0                             # punct_burstiness
]
return vec
```

APPENDIX 2: SAMPLE BENCHMARK RECORDS

Table XXXI lists sample records from the benchmark with all three labels, two per language and claim type.

TABLE XXXI
SAMPLE BENCHMARK RECORDS

ID	Language	Statement (truncated)	Ver.	Hate	Ambiguity
TCF_00001	Hinglish	Social media par khabar hai ki new digital tax rules me kuch badlav ho sakta hai, halanki official confirmation ka intez…	Real	Non-Hateful	Borderline
TCF_00002	English	The Peer-reviewed Medical Journal has issued official regulatory guidance on clean energy policy during its scheduled re…	Real	Non-Hateful	Clear Fact
TCF_00003	Tamil	அதிர்ச்சி செய்தி!! ரகசிய உண்மை வெளியானது!! மழை வெள்ள முன்னெச்சரிக்கை மக்களை ஏமாற்றும் மிகப்பெரிய சதி! உடனடியாக வாட்ஸ்அப்…	Fake	Non-Hateful	Sensational Deception
TCF_00005	Tanglish	TNPSC Official Board has officially announced that bus fare subsidy official website-la formal notification update panni…	Real	Non-Hateful	Clear Fact
TCF_00006	Hinglish	Supreme Court Public Relations Office ne 500 rupee notes ke liye new guidelines official portal pe upload kar diya hai.	Real	Non-Hateful	Clear Fact
TCF_00007	English	MUST SHARE BEFORE DELETED: Clean Energy Policy poisons local groundwater. Share before authorities delete this link!!!	Fake	Non-Hateful	Sensational Deception
TCF_00010	Tanglish	BAYANGARAMAANA UNMAI: Doctors hide panraanga!! Electricity Tariff Revision secret chip insert panraanga nu doctor reveal…	Fake	Non-Hateful	Sensational Deception
TCF_00013	Tamil	தமிழக சுகாதாரத்துறை புதிய தடுப்பூசி தொடர்பாக அதிகாரப்பூர்வ செய்திக்குறிப்பு இன்று வெளியிடப்பட்டுள்ளது.	Real	Non-Hateful	Clear Fact
TCF_00017	Hinglish	KHATARNAK SACH EXPOSED!! Dekhiye ye viral sach!! Vaccination Drive aaj raat se illegal declare hone wala hai. Jaldi se j…	Fake	Hateful	Sensational Deception
TCF_00023	English	Unconfirmed reports suggest officials may consider revising parameters regarding 5G telecom network, though formal docum…	Real	Non-Hateful	Borderline
TCF_00032	Hindi	भारतीय रिजर्व बैंक (RBI) ने आज एक आधिकारिक प्रेस विज्ञप्ति जारी कर नई शिक्षा नीति की प्रगति रिपोर्ट प्रस्तुत की है।	Real	Non-Hateful	Clear Fact
TCF_00039	Hindi	वायरल अलर्ट: स्वास्थ्य विभाग ने छिपाई यह गंभीर बात: राष्ट्रीय स्वास्थ्य मिशन को लेकर अधिकारियों पर लगे बड़े आरोप। इस संद…	Fake	Non-Hateful	Sensational Deception
TCF_00040	Tamil	மழை வெள்ள முன்னெச்சரிக்கை தொடர்பாக அடுத்த வாரம் புதிய மாற்றம் வர வாய்ப்பு உள்ளதாக கூறப்படுகிறது, ஆனால் அதிகாரப்பூர்வ உறு…	Real	Non-Hateful	Borderline
TCF_00087	Tanglish	Oruvela new tech park project patriya decision adutha month change aaga chance iruku nu news kelvipatten, but unmaiya nu…	Real	Non-Hateful	Borderline

APPENDIX 3: SAMPLE TEST PREDICTIONS

Table XXXII lists a stratified sample of test predictions with the true label, the fake probability, the confidence g(x), the final decision at the calibrated threshold and the hate probability.

TABLE XXXII
SAMPLE TEST PREDICTIONS

Lang.	Claim (truncated)	True	P(fake)	g(x)	Decision	P(hate)
English	Unconfirmed reports suggest officials may consider revising parameters regarding 5G telecom network, thou…	Real	0.460	0.178	NOT SURE	0.001
English	The Indian Council of Medical Research (ICMR) has published quarterly clinical trial metrics clean energy…	Real	0.000	1.000	REAL	0.000
English	EXPOSED: Massive fraud in health system: Covid Booster Shot causes permanent cell mutation. Share before …	Fake	1.000	1.000	FAKE	0.481

Lang.	Claim (truncated)	True	P(fake)	g(x)	Decision	P(hate)
Hindi	स्वास्थ्य एवं परिवार कल्याण मंत्रालय ने आज एक आधिकारिक प्रेस विज्ञप्ति जारी कर डिजिटल मुद्रा दिशानिर्देश …	Real	0.000	1.000	REAL	0.000
Hindi	सनसनीखेज खुलासा!! इस बड़े घोटाले की सच्चाई आई सामने: डिजिटल मुद्रा दिशानिर्देश को लेकर अधिकारियों पर लगे …	Fake	1.000	1.000	FAKE	0.007
Hinglish	Social media par khabar hai ki free ration scheme me kuch badlav ho sakta hai, halanki official confirmat…	Real	0.362	0.195	NOT SURE	0.001
Hinglish	Supreme Court Public Relations Office ne 500 rupee notes ke liye new guidelines official portal pe upload…	Real	0.000	1.000	REAL	0.000
Hinglish	KHATARNAK SACH EXPOSED!! Dekhiye ye viral sach!! 500 Rupee Notes poora public ko bewaqoof banane ki saazi…	Fake	1.000	1.000	FAKE	0.997
Tamil	பள்ளி மாணவர் நலத்திட்டம் தொடர்பாக அடுத்த வாரம் புதிய மாற்றம் வர வாய்ப்பு உள்ளதாக கூறப்படுகிறது, ஆனால் அதி…	Fake	0.376	0.137	NOT SURE	0.001
Tamil	தமிழ்நாடு அரசு தேர்வுத்துறை மழை வெள்ள முன்னெச்சரிக்கை தொடர்பாக அதிகாரப்பூர்வ செய்திக்குறிப்பு இன்று வெளிய…	Real	0.001	1.000	REAL	0.000
Tamil	அதிர்ச்சி செய்தி!! ரகசிய உண்மை வெளியானது!! மழை வெள்ள முன்னெச்சரிக்கை மக்களை ஏமாற்றும் மிகப்பெரிய சதி! உடன…	Fake	0.999	1.000	FAKE	0.003
Tanglish	Oruvela new tech park project patriya decision adutha month change aaga chance iruku nu news kelvipatten,…	Real	0.478	0.121	NOT SURE	0.001
Tanglish	District Collectorate Office has officially announced that new tech park project successfully review pann…	Real	0.000	1.000	REAL	0.000
Tanglish	VIRAL ALERT: Secrete conspiracy exposed!! New Tech Park Project dangerous poison irukku nu proof vandhach…	Fake	1.000	1.000	FAKE	0.997

APPENDIX 4: COMPLETE MULTILINGUAL LEXICONS

Table XXXIII lists every entry of the eight lexicons used by the stylometric stream, exactly as implemented in features/stylometric.py. Entries are matched against lower-cased tokens produced by the script-aware tokenizer.

TABLE XXXIII
COMPLETE LEXICON ENTRIES

Lexicon	Entries
Hedges	afwaah, allegedly, apparently, bolte, claimed, could, doubt-ah, kahate, kelvipatten, lagta, maybe, might, oruvela, perhaps, possibly, purportedly, reportedly, rumor, rumour, seems, shayad, solraanga, suggests, suna, supposedly, unconfirmed, अफवाह, शायद, ஒருவேளை, கூறப்படுகிறது, சொல்கிறார்கள், வதந்தி
Sensational	adhirchi, banned, bayangarama, bhasad, bombshell, breaking, breaking-la, destroy, dhamaal, explosive, exposed, hoax, hungama, insane, khatarnak, khulasa, marmam, miracle, outrageous, secret, sharmnak, shocking, terrifying, unbelievable, urgent, viral-ah, खुलासा, धमाका, सनसनीखेज, हंगामा, அதிர்ச்சி, சதி, பரபரப்பு, மர்மம், வெளியானது
Superlatives	always, best, biggest, ekdam, ever, greatest, least, most, mutrilum, never, perfect, periya, sabse, smallest, ultimate, worst, सबसे, மிகப்பெரிய, முதல்முறை, முழுவதும்
Negations	illa, ille, jhooth, mat, n't, nahi, neither, never, nhi, no, nobody, none, not, nothing, nowhere, pannatheenga, poi, vendam, झूठ, नहीं, मत, இல்லை, கூடாது, பொய், வேண்டாம்
First person	enaku, hamara, hum, i, main, me, mera, mine, my, naan, namba, our, ours, us, we, நமது, நான், நாம்
Second person	aap, apka, nee, neenga, tera, tu, tum, ungala, you, your, yours, உன், நீ, நீங்கள்
Modals	can, could, may, might, must, shall, should, will, would
Positive affect	amazing, azhagu, badhiya, great, nalladhu, proven, safe, shandar, success, super, trusted, win, wonderful, बढ़िया, शानदार, அழகு, நன்மை, வெற்றி
Negative affect	attack, barbaad, corrupt, crisis, danger, death, dhokha, fear, fraud, hate, jhooth, kill, kolai, lie, mosadi, panic, threat, thurogi, toxic, खतरा, धोखा, मौत, அபாயம், கொலை, துரோகம், மோசடி

APPENDIX 5: BENCHMARK TEMPLATE POOLS

Table XXXIV summarises the template pools in data/generate_dataset.py from which the benchmark claims are composed. Each sensational claim joins an opener, a topic and a fabricated consequence; each clear fact joins an official source, a topic and a measured verb phrase; borderline claims use a fixed rumour frame per language.

TABLE XXXIV
TEMPLATE POOLS USED TO GENERATE THE BENCHMARK

Language	Sensational openers (examples)	Official sources (examples)	Topics (examples)
English	SHOCKING!! Scientists EXPOSED; URGENT ALERT: Truth they hide; MUST SHARE BEFORE DELETED	World Health Organization; ICMR; Reuters Fact Check Desk; National Statistics Bureau	COVID booster shot; 5G telecom network; central bank digital rupee
Tamil	அதிர்ச்சி செய்தி!! ரகசிய உண்மை வெளியானது; பரபரப்பு தகவல்! உடனே பாருங்கள்	தமிழக சுகாதாரத்துறை; சென்னை உயர்நீதிமன்றம்; அண்ணா பல்கலைக்கழகம்	புதிய தடுப்பூசி; ரேஷன் அட்டை புதிய விதி; மழை வெள்ள முன்னெச்சரிக்கை
Tanglish	ADHIROOPAM BREAKING!!; VIRAL ALERT: Secrete conspiracy exposed; SHOCKING LEAK	Tamil Nadu Health Department; Anna University Exam Controller; TNPSC Official Board	semester exam schedule; bus fare subsidy; electricity tariff revision
Hinglish	KHATARNAK SACH EXPOSED!!; SABSE BADA SCAM LEAK; SAVDHAN!! Secret circular viral	Reserve Bank of India; Health Ministry Official Portal; Railway Recruitment Board	500 rupee notes; railway exam schedule; free ration scheme
Hindi	सनसनीखेज खुलासा!! इस बड़े घोटाले की सच्चाई आई सामने; वायरल अलर्ट	स्वास्थ्य एवं परिवार कल्याण मंत्रालय; भारतीय रिजर्व बैंक; नीति आयोग	राष्ट्रीय स्वास्थ्य मिशन; डिजिटल मुद्रा दिशानिर्देश; नई शिक्षा नीति

APPENDIX 6: GLOSSARY

TABLE XXXV
GLOSSARY OF KEY TERMS

Term	Meaning in this report
Abstention	Returning "Not Sure" instead of a label when confidence g(x) is below τ.
Borderline claim	An unconfirmed rumour whose veracity cannot be determined from its text; labelled at random in the benchmark.
Code-mixing	Use of two languages within one sentence, such as Tamil and English in Tanglish.
Coverage	Share of claims the model answers rather than defers.
Gate weights (α)	Per-claim weights that decide how much each evidence stream contributes.
Grouped split	A data split in which all copies of the same text fall into one split.
Occlusion saliency	Change in P(fake) when one word is removed; shows which words drive a verdict.
Risk–coverage curve	Error on answered claims plotted against coverage.
Selective risk	Error rate on the claims the model answers.
SIR model	Susceptible–Infected–Recovered model of how a message spreads through a population.
Stylometry	Measurement of writing-style features such as punctuation, casing and word choice.
Transliteration	Writing a language in another script, usually Roman letters.

APPENDIX 7: USER MANUAL

1. Install Python 3.10 or later, then run "pip install -r requirements.txt" in the project folder.
2. Start the application with "python app.py"; the first start downloads and loads the MuRIL encoder.
3. Open http://localhost:7860 in a browser.
4. In Tab 1, paste a claim in English, Tamil, Tanglish, Hinglish or Hindi, optionally move the coverage slider, and press "Analyze Article with TCF-Net".
5. Read the veracity badge (REAL, FAKE or NOT SURE), the hate badge, the confidence gauge, the modality donut and the word-saliency chart.
6. In Tab 2, paste the same claim and press "Extract Stylometric & Cascade Features" to see its stylometric profile and simulated spread.
7. In Tab 3, move the coverage slider to see how the error rate changes and read the abstention audit tables.
8. To retrain, run "python train.py --backbone google/muril-base-cased"; to regenerate figures, run "python -m evaluation.make_figures".

APPENDIX 8: REPRODUCIBILITY DETAILS

All results in this report were produced by "python train.py --backbone google/muril-base-cased" with seed 42 on the bundled benchmark and regenerated twice with identical metrics. Artifacts written to the artifacts folder include the checkpoint (tcfnet.pt), the fitted scalers and projectors (scalers.joblib), final_metrics.json, per_language_metrics.csv, per_ambiguity_metrics.csv, baseline_comparison.csv, risk_coverage_curve.csv, training_history.csv and test_predictions.csv with per-claim probabilities, confidences and gate weights. The MuRIL embeddings are cached so that repeated runs skip the encoder; with the cache, a full run including baselines and ablations takes a few minutes on a laptop CPU.

APPENDIX 9: ADDITIONAL SCREENSHOTS

Figs. 22–24 show the three tabs of the application on English and Hinglish input, as captured from the running system.



Fig. 22. Tab 1 on a Hinglish fake and hateful claim (full page).

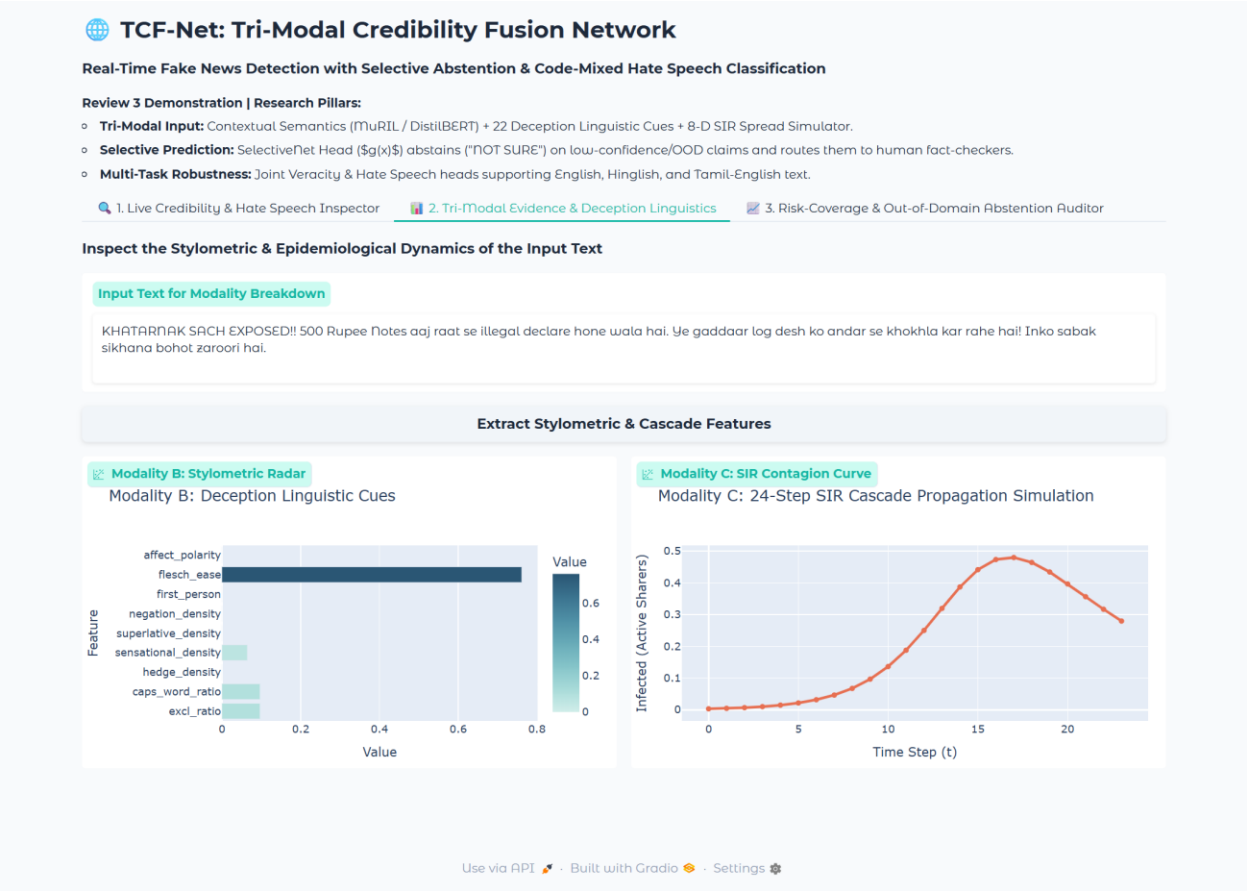


Fig. 23. Tab 2 on a Hinglish claim (full page).

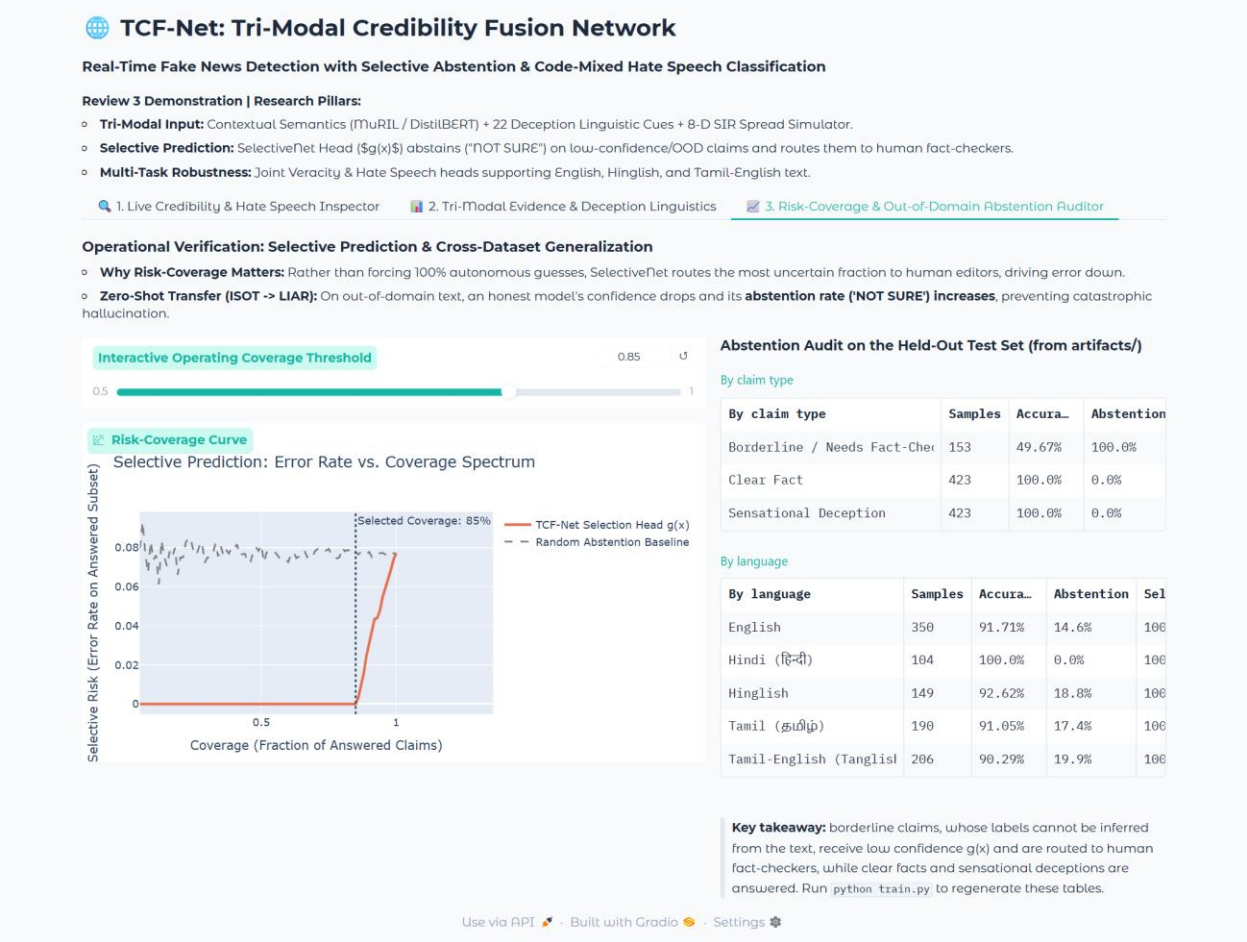


Fig. 24. Tab 3 risk–coverage auditor (full page).

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